

SOLIDWORKS API (.NET VBA C#) — Reference (120/337): IFeatureManager (part 7 of 9)

SOLIDWORKS API Documentation (converted from CHM)

IFeatureManager (part 7 of 9) (Type)

SOLIDWORKS API Help

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InsertProtrusionSwept Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertProtrusionSwept Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::InsertProtusionSwept3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertProtrusionSwept( _  
    ByVal Propagate As System.Boolean, _  
    ByVal Alignment As System.Boolean, _  
    ByVal TwistCtrlOption As System.Short, _  
    ByVal KeepTangency As System.Boolean, _  
    ByVal ForceNonRational As System.Boolean, _  
    ByVal StartMatchingType As System.Short, _  
    ByVal EndMatchingType As System.Short, _  
    ByVal IsThinBody As System.Boolean, _  
    ByVal Thickness1 As System.Double, _  
    ByVal Thickness2 As System.Double, _  
    ByVal ThinType As System.Short, _  
    ByVal Merge As System.Boolean, _  
    ByVal UseFeatScope As System.Boolean, _  
    ByVal UseAutoSelect As System.Boolean _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Propagate As System.Boolean  
Dim Alignment As System.Boolean  
Dim TwistCtrlOption As System.Short  
Dim KeepTangency As System.Boolean  
Dim ForceNonRational As System.Boolean  
Dim StartMatchingType As System.Short  
Dim EndMatchingType As System.Short  
Dim IsThinBody As System.Boolean  
Dim Thickness1 As System.Double  
Dim Thickness2 As System.Double  
Dim ThinType As System.Short  
Dim Merge As System.Boolean  
Dim UseFeatScope As System.Boolean  
Dim UseAutoSelect As System.Boolean  
Dim value As Feature
```

```
value = instance.InsertProtrusionSwept(Propagate, Alignment, TwistCtrlOption, KeepTangency,  
ForceNonRational, StartMatchingType, EndMatchingType, IsThinBody, Thickness1, Thickness2,  
ThinType, Merge, UseFeatScope, UseAutoSelect)
```

C#

```
Feature InsertProtrusionSwept(  
    System.bool Propagate,  
    System.bool Alignment,  
    System.short TwistCtrlOption,  
    System.bool KeepTangency,  
    System.bool ForceNonRational,  
    System.short StartMatchingType,  
    System.short EndMatchingType,  
    System.bool IsThinBody,  
    System.double Thickness1,  
    System.double Thickness2,  
    System.short ThinType,
```

```
System.bool Merge,  
System.bool UseFeatScope,  
System.bool UseAutoSelect  
)
```

C++/CLI

```
Feature^ InsertProtrusionSwept(  
& System.bool Propagate,  
& System.bool Alignment,  
& System.short TwistCtrlOption,  
& System.bool KeepTangency,  
& System.bool ForceNonRational,  
& System.short StartMatchingType,  
& System.short EndMatchingType,  
& System.bool IsThinBody,  
& System.double Thickness1,  
& System.double Thickness2,  
& System.short ThinType,  
& System.bool Merge,  
& System.bool UseFeatScope,  
& System.bool UseAutoSelect  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Propagate

Alignment

TwistCtrlOption

KeepTangency

ForceNonRational

StartMatchingType

EndMatchingType

IsThinBody

Thickness1

Thickness2

ThinType

Merge

UseFeatScope

UseAutoSelect

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertProtrusionSwept](#).

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

SOLIDWORKS API Help

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InsertProtrusionSwept2 Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertProtrusionSwept2 Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager.InsertProtusionSwept3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertProtrusionSwept2( _  
    ByVal Propagate As System.Boolean, _
```

```

    ByVal Alignment As System.Boolean, _
    ByVal TwistCtrlOption As System.Short, _
    ByVal KeepTangency As System.Boolean, _
    ByVal ForceNonRational As System.Boolean, _
    ByVal StartMatchingType As System.Short, _
    ByVal EndMatchingType As System.Short, _
    ByVal IsThinBody As System.Boolean, _
    ByVal Thickness1 As System.Double, _
    ByVal Thickness2 As System.Double, _
    ByVal ThinType As System.Short, _
    ByVal PathAlign As System.Short, _
    ByVal Merge As System.Boolean, _
    ByVal UseFeatScope As System.Boolean, _
    ByVal UseAutoSelect As System.Boolean _
) As Feature

```

Visual Basic (Usage)



```

Dim instance As IFeatureManager
Dim Propagate As System.Boolean
Dim Alignment As System.Boolean
Dim TwistCtrlOption As System.Short
Dim KeepTangency As System.Boolean
Dim ForceNonRational As System.Boolean
Dim StartMatchingType As System.Short
Dim EndMatchingType As System.Short
Dim IsThinBody As System.Boolean
Dim Thickness1 As System.Double
Dim Thickness2 As System.Double
Dim ThinType As System.Short
Dim PathAlign As System.Short
Dim Merge As System.Boolean
Dim UseFeatScope As System.Boolean
Dim UseAutoSelect As System.Boolean
Dim value As Feature

```

```

value = instance.InsertProtrusionSwept2(Propagate, Alignment, TwistCtrlOption, KeepTangency,
ForceNonRational, StartMatchingType, EndMatchingType, IsThinBody, Thickness1, Thickness2,
ThinType, PathAlign, Merge, UseFeatScope, UseAutoSelect)

```

C#

```

Feature InsertProtrusionSwept2(
    System.bool Propagate,
    System.bool Alignment,
    System.short TwistCtrlOption,
    System.bool KeepTangency,
    System.bool ForceNonRational,
    System.short StartMatchingType,
    System.short EndMatchingType,
    System.bool IsThinBody,
    System.double Thickness1,
    System.double Thickness2,
    System.short ThinType,
    System.short PathAlign,
    System.bool Merge,
    System.bool UseFeatScope,
    System.bool UseAutoSelect
)

```

C++/CLI

```

Feature^ InsertProtrusionSwept2(
    & System.bool Propagate,
    & System.bool Alignment,
    & System.short TwistCtrlOption,
    & System.bool KeepTangency,
    & System.bool ForceNonRational,
    & System.short StartMatchingType,
    & System.short EndMatchingType,

```

```

& System.bool IsThinBody,
& System.double Thickness1,
& System.double Thickness2,
& System.short ThinType,
& System.short PathAlign,
& System.bool Merge,
& System.bool UseFeatScope,
& System.bool UseAutoSelect
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Propagate

Alignment

TwistCtrlOption

KeepTangency

ForceNonRational

StartMatchingType

EndMatchingType

IsThinBody

Thickness1

Thickness2

ThinType

PathAlign

Merge

UseFeatScope

UseAutoSelect

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertProtrusionSwept2](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

SOLIDWORKS API Help

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InsertProtrusionSwept3 Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : [InsertProtrusionSwept3 Method \(IFeatureManager\)](#)

Obsolete. Superseded by [IFeatureManager::InsertProtrusionSwept4](#).

.NET Syntax

Visual Basic (Declaration)

```

Function InsertProtrusionSwept3( _
    ByVal Propagate As System.Boolean, _
    ByVal Alignment As System.Boolean, _
    ByVal TwistCtrlOption As System.Short, _
    ByVal KeepTangency As System.Boolean, _
    ByVal AdvancedSmoothing As System.Boolean, _
    ByVal StartMatchingType As System.Short, _
    ByVal EndMatchingType As System.Short, _
    ByVal IsThinBody As System.Boolean, _
    ByVal Thickness1 As System.Double, _
    ByVal Thickness2 As System.Double, _
    ByVal ThinType As System.Short, _
    ByVal PathAlign As System.Short, _
    ByVal Merge As System.Boolean, _

```

```

    ByVal UseFeatScope As System.Boolean, _
    ByVal UseAutoSelect As System.Boolean, _
    ByVal TwistAngle As System.Double, _
    ByVal BMergeSmoothFaces As System.Boolean _
) As Feature

```

Visual Basic (Usage)



```

Dim instance As IFeatureManager
Dim Propagate As System.Boolean
Dim Alignment As System.Boolean
Dim TwistCtrlOption As System.Short
Dim KeepTangency As System.Boolean
Dim BAdvancedSmoothing As System.Boolean
Dim StartMatchingType As System.Short
Dim EndMatchingType As System.Short
Dim IsThinBody As System.Boolean
Dim Thickness1 As System.Double
Dim Thickness2 As System.Double
Dim ThinType As System.Short
Dim PathAlign As System.Short
Dim Merge As System.Boolean
Dim UseFeatScope As System.Boolean
Dim UseAutoSelect As System.Boolean
Dim TwistAngle As System.Double
Dim BMergeSmoothFaces As System.Boolean
Dim value As Feature

```

```

value = instance.InsertProtrusionSwept3(Propagate, Alignment, TwistCtrlOption, KeepTangency,
BAdvancedSmoothing, StartMatchingType, EndMatchingType, IsThinBody, Thickness1, Thickness2,
ThinType, PathAlign, Merge, UseFeatScope, UseAutoSelect, TwistAngle, BMergeSmoothFaces)

```

C#

```

Feature InsertProtrusionSwept3(
    System.bool Propagate,
    System.bool Alignment,
    System.short TwistCtrlOption,
    System.bool KeepTangency,
    System.bool BAdvancedSmoothing,
    System.short StartMatchingType,
    System.short EndMatchingType,
    System.bool IsThinBody,
    System.double Thickness1,
    System.double Thickness2,
    System.short ThinType,
    System.short PathAlign,
    System.bool Merge,
    System.bool UseFeatScope,
    System.bool UseAutoSelect,
    System.double TwistAngle,
    System.bool BMergeSmoothFaces
)

```

C++/CLI

```

Feature^ InsertProtrusionSwept3(
& System.bool Propagate,
& System.bool Alignment,
& System.short TwistCtrlOption,
& System.bool KeepTangency,
& System.bool BAdvancedSmoothing,
& System.short StartMatchingType,
& System.short EndMatchingType,
& System.bool IsThinBody,
& System.double Thickness1,
& System.double Thickness2,
& System.short ThinType,
& System.short PathAlign,
& System.bool Merge,

```

```
& System.bool UseFeatScope,  
& System.bool UseAutoSelect,  
& System.double TwistAngle,  
& System.bool BMergeSmoothFaces  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Propagate

True propagates the swept protrusion to the next tangent edge, false does not

Alignment

True causes the swept protrusion to go through the end faces if the curve used for the sweep goes from one face to another or from one edge to another, false causes the swept protrusion to begin and end perpendicular to the sweep curve and it cannot break through the two end faces of the body

TwistCtrlOption

Twist control options as defined in [swTwistControlType_e](#)

KeepTangency

If the sweep section has tangent segments, then true to cause the corresponding surfaces in the resulting sweep to be tangent, false to not

BAdvancedSmoothing

If the sweep section has circular or elliptical arcs, then true to approximate the sections and smooth the surfaces, false to not

StartMatchingType

Tangency type as defined in [swTangencyType_e](#)

EndMatchingType

Tangency type as defined in [swTangencyType_e](#)

IsThinBody

True if this feature is a thin body, false if not

Thickness1

Thickness value for the first direction

Thickness2

Thickness value for the second direction

ThinType

Thin wall type as defined in [swThinWallType_e](#)

PathAlign

Align path type (see **Remarks**)

Merge

True to merge the results in a multibody part, false to not

UseFeatScope

True if the feature only affects selected bodies, false if the feature affects all bodies

UseAutoSelect

True to automatically select all bodies and have the feature affect those bodies, false to select the bodies the feature affects (see **Remarks**)

TwistAngle

If TwistCtrlOption set to swTwistControlType_e.swTwistControlConstantTwistAlongPath, then specify end twist angle

BMergeSmoothFaces

True to merge smooth faces, false to not

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertProtrusionSwept3](#).

Remarks

Use [IModelDocExtension::SelectByID2](#) to select the profile and sweep curves. The mark for:

- 1 = Profile selection
- 2 = Guide curve selection, if provided
- 4 = Sweep path

The PathAlign argument is available when TwistCtrlOption is set to 0 (follow path) and can take one of these values:

- 0 = None; no correction (default)
- 2 = Direction vector; a plane, planar face, or line defines the path
- 3 = All faces; includes neighboring faces

When UseAutoSelect is false, the user must select the bodies that the feature will affect.

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

[ISweepFeatureData Interface](#)

[IFeatureManager::InsertCutSwept3 Method](#)

[IFeatureManager::InsertSweepSurface2 Method](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

Obsolete. See [Remarks](#).

Visual Basic Syntax

Visual Basic (Declaration)

```

Function InsertProtrusionSwept4( _
    ByVal Propagate As System.Boolean, _
    ByVal Alignment As System.Boolean, _
    ByVal TwistCtrlOption As System.Integer, _
    ByVal KeepTangency As System.Boolean, _
    ByVal BAdvancedSmoothing As System.Boolean, _
    ByVal StartMatchingType As System.Integer, _
    ByVal EndMatchingType As System.Integer, _
    ByVal IsThinBody As System.Boolean, _
    ByVal Thickness1 As System.Double, _
    ByVal Thickness2 As System.Double, _
    ByVal ThinType As System.Integer, _
    ByVal PathAlign As System.Integer, _
    ByVal Merge As System.Boolean, _
    ByVal UseFeatScope As System.Boolean, _
    ByVal UseAutoSelect As System.Boolean, _
    ByVal TwistAngle As System.Double, _
    ByVal BMergeSmoothFaces As System.Boolean, _
    ByVal CircularProfile As System.Boolean, _
    ByVal CircularProfileDiameter As System.Double, _
    ByVal Direction As System.Integer _
) As Feature

```

Visual Basic (Usage)



```

Dim instance As IFeatureManager
Dim Propagate As System.Boolean
Dim Alignment As System.Boolean
Dim TwistCtrlOption As System.Integer
Dim KeepTangency As System.Boolean
Dim BAdvancedSmoothing As System.Boolean
Dim StartMatchingType As System.Integer
Dim EndMatchingType As System.Integer
Dim IsThinBody As System.Boolean
Dim Thickness1 As System.Double
Dim Thickness2 As System.Double
Dim ThinType As System.Integer
Dim PathAlign As System.Integer
Dim Merge As System.Boolean
Dim UseFeatScope As System.Boolean
Dim UseAutoSelect As System.Boolean
Dim TwistAngle As System.Double
Dim BMergeSmoothFaces As System.Boolean
Dim CircularProfile As System.Boolean
Dim CircularProfileDiameter As System.Double
Dim Direction As System.Integer
Dim value As Feature

```

```

value = instance.InsertProtrusionSwept4(Propagate, Alignment, TwistCtrlOption, KeepTangency,
BAdvancedSmoothing, StartMatchingType, EndMatchingType, IsThinBody, Thickness1, Thickness2,
ThinType, PathAlign, Merge, UseFeatScope, UseAutoSelect, TwistAngle, BMergeSmoothFaces,
CircularProfile, CircularProfileDiameter, Direction)

```

C#

```

Feature InsertProtrusionSwept4(
    System.bool Propagate,
    System.bool Alignment,
    System.int TwistCtrlOption,
    System.bool KeepTangency,
    System.bool BAdvancedSmoothing,
    System.int StartMatchingType,

```

```

System.int EndMatchingType,
System.bool IsThinBody,
System.double Thickness1,
System.double Thickness2,
System.int ThinType,
System.int PathAlign,
System.bool Merge,
System.bool UseFeatScope,
System.bool UseAutoSelect,
System.double TwistAngle,
System.bool BMergeSmoothFaces,
System.bool CircularProfile,
System.double CircularProfileDiameter,
System.int Direction
)

```

C++/CLI

```

Feature^ InsertProtrusionSwept4(
& System.bool Propagate,
& System.bool Alignment,
& System.int TwistCtrlOption,
& System.bool KeepTangency,
& System.bool BAdvancedSmoothing,
& System.int StartMatchingType,
& System.int EndMatchingType,
& System.bool IsThinBody,
& System.double Thickness1,
& System.double Thickness2,
& System.int ThinType,
& System.int PathAlign,
& System.bool Merge,
& System.bool UseFeatScope,
& System.bool UseAutoSelect,
& System.double TwistAngle,
& System.bool BMergeSmoothFaces,
& System.bool CircularProfile,
& System.double CircularProfileDiameter,
& System.int Direction
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Propagate

True propagates the swept protrusion to the next tangent edge, false does not

Alignment

True causes the swept protrusion to go through the end faces if the curve used for the sweep goes from one face to another or from one edge to another, false causes the swept protrusion to begin and end perpendicular to the sweep curve; thus, it cannot break through the two end faces of the body

TwistCtrlOption

Twist control options as defined in [swTwistControlType_e](#)

KeepTangency

If the sweep section has tangent segments, then true to cause the corresponding surfaces in the resulting sweep to be tangent, false to not

BAdvancedSmoothing

If the sweep section has circular or elliptical arcs, then true to approximate the sections and smooth the surfaces, false to not

StartMatchingType

Tangency type as defined in [swTangencyType_e](#)

EndMatchingType

Tangency type as defined in [swTangencyType_e](#)

IsThinBody

True if this feature is a thin body, false if not

Thickness1

Thickness value for the first direction

Thickness2

Thickness value for the second direction

ThinType

Thin wall type as defined in [swThinWallType_e](#)

PathAlign

Align path type (see **Remarks**)

Merge

True to merge the results in a multibody part, false to not

UseFeatScope

True if the feature only affects selected bodies, false if the feature affects all bodies

UseAutoSelect

True to automatically select all bodies and have the feature affect those bodies, false to select the bodies the feature affects (see **Remarks**)

TwistAngle

If *TwistCtrlOption* is *swTwistControlType_e.swTwistControlConstantTwistAlongPath*, then specify end twist angle

BMergeSmoothFaces

True to merge smooth faces, false to not

CircularProfile

True to use a circular profile, false to use the selected sketch profile or solid body

CircularProfileDiameter
If CircularProfile is true, then specify the diameter of the circular profile
Direction
Direction as defined in [swSweepDirection_e](#) (see **Remarks**)

Return Value

[Feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertProtrusionSwept4](#).

Example

[Get Guide Curves in Sweep Feature \(C#\)](#)
[Get Guide Curves in Sweep Feature \(VB.NET\)](#)
[Get Guide Curves in Sweep Feature \(VBA\)](#)
[Select Multiple Sketches for Sweep Path \(C#\)](#)
[Select Multiple Sketches for Sweep Path \(VB.NET\)](#)
[Select Multiple Sketches for Sweep Path \(VBA\)](#)

Remarks

SOLIDWORKS 2018 introduces a new sweep architecture, making this method obsolete. See [Sweep Features and SweepFeatureData Objects](#) to create this swept-boss or base feature.

Use [IModelDocExtension::SelectByID2](#) to select the sketch profile and sweep curves. Specify these marks:

- 1 = Sketch profile
- 2 = Guide curve, if provided
- 4 = Sweep path

The PathAlign argument is available when TwistCtrlOption is set to swTwistControlType_e.swTwistControlFollowPath and can take one of these values:

- 0 = None; no correction (default)
- 2 = Direction vector; a plane, planar face, or line defines the path
- 3 = All faces; includes neighboring faces

When UseAutoSelect is false, the user must select the bodies that the feature will affect.

Direction only applies to sketch profiles and only when the sketch profile is not coincident with an end of the path.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeatureManager::InsertCutSwept5 Method \(\)](#)
[IFeatureManager::InsertSweepSurface3 Method \(\)](#)
[ISweepFeatureData Interface](#)

Availability

SOLIDWORKS 2017 FCS, Revision Number 25.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertReferencePoint Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertReferencePoint Method (IFeatureManager)

Creates the geometry for the reference points based on any of these selected entities: edges, faces, planes, vertices, or sketch geometry.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertReferencePoint( _  
    ByVal NRefPointType As System.Integer, _  
    ByVal NRefPointAlongCurveType As System.Integer, _
```

```

    ByVal DDistance\_or\_Percent As System.Double, _
    ByVal NumberOfRefPoints As System.Integer _
) As System.Object

```

Visual Basic (Usage)



```

Dim instance As IFeatureManager
Dim NRefPointType As System.Integer
Dim NRefPointAlongCurveType As System.Integer
Dim DDistance_or_Percent As System.Double
Dim NumberOfRefPoints As System.Integer
Dim value As System.Object

value = instance.InsertReferencePoint(NRefPointType, NRefPointAlongCurveType,
DDistance_or_Percent, NumberOfRefPoints)

```

C#

```

System.object InsertReferencePoint(
    System.int NRefPointType,
    System.int NRefPointAlongCurveType,
    System.double DDistance\_or\_Percent,
    System.int NumberOfRefPoints
)

```

C++/CLI

```

System.Object^ InsertReferencePoint(
    & System.int NRefPointType,
    & System.int NRefPointAlongCurveType,
    & System.double DDistance\_or\_Percent,
    & System.int NumberOfRefPoints
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NRefPointType

Type of reference point as defined by [swRefPointType_e](#)

NRefPointAlongCurveType

Distance, percentage, or evenly distributed as defined by [swRefPointAlongCurveType_e](#)

DDistance_or_Percent

Distance at which to create the reference point on the selected entities or percentage of the length of the selected entities at which to create the reference point if *NRefPointAlongCurveType* is *swRefPointAlongCurveDistance* or *swRefPointAlongCurvePercentage*, respectively

NumberOfRefPoints

Number of reference points to create and evenly distribute on the selected entities if *swRefPointAlongCurveType* is *swRefPointAlongCurveEvenlyDistributed*

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertReferencePoint](#).

Example

[Insert Reference Points \(VBA\)](#)

Remarks

This method creates one or more reference point features and adds them to the FeatureManager design tree. If the reference point feature is not created, a NULL value is returned.

The NumberOfRefPoints argument must contain a value of 1 to successfully create one reference point feature.

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

[IFeatureManager::InsertReferencePoint Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertRefPlane Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertRefPlane Method (IFeatureManager)

Inserts a constraint-based reference plane using the selected reference entities.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertRefPlane( _  
    ByVal FirstConstraint As System.Integer, _  
    ByVal FirstConstraintAngleOrDistance As System.Double, _  
    ByVal SecondConstraint As System.Integer, _  
    ByVal SecondConstraintAngleOrDistance As System.Double, _  
    ByVal ThirdConstraint As System.Integer, _  
    ByVal ThirdConstraintAngleOrDistance As System.Double _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim FirstConstraint As System.Integer  
Dim FirstConstraintAngleOrDistance As System.Double  
Dim SecondConstraint As System.Integer  
Dim SecondConstraintAngleOrDistance As System.Double  
Dim ThirdConstraint As System.Integer  
Dim ThirdConstraintAngleOrDistance As System.Double  
Dim value As System.Object
```

```
value = instance.InsertRefPlane(FirstConstraint, FirstConstraintAngleOrDistance, SecondConstraint,  
SecondConstraintAngleOrDistance, ThirdConstraint, ThirdConstraintAngleOrDistance)
```

C#

```
System.object InsertRefPlane(  
    System.int FirstConstraint,  
    System.double FirstConstraintAngleOrDistance,  
    System.int SecondConstraint,  
    System.double SecondConstraintAngleOrDistance,  
    System.int ThirdConstraint,  
    System.double ThirdConstraintAngleOrDistance  
)
```

C++/CLI

```
System.Object^ InsertRefPlane(  
& System.int FirstConstraint,  
& System.double FirstConstraintAngleOrDistance,  
& System.int SecondConstraint,  
& System.double SecondConstraintAngleOrDistance,  
& System.int ThirdConstraint,  
& System.double ThirdConstraintAngleOrDistance  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FirstConstraint

First constraint as defined in [swRefPlaneReferenceConstraints](#) e

FirstConstraintAngleOrDistance

Angle or distance of the first constraint

SecondConstraint

Second constraint as defined in [swRefPlaneReferenceConstraints_e](#)
SecondConstraintAngleOrDistance
Angle or distance of the second constraint
ThirdConstraint
Third constraint as defined in [swRefPlaneReferenceConstraints_e](#)
ThirdConstraintAngleOrDistance
Angle or distance of the third constraint

Return Value

[Reference plane](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertRefPlane](#).

Example

[Insert Reference Plane \(C#\)](#)
[Insert Reference Plane \(VB.NET\)](#)
[Insert Reference Plane \(VBA\)](#)
[Insert Fill-surface Feature \(C#\)](#)
[Insert Fill-surface Feature \(VB.NET\)](#)
[Insert Fill-surface Feature \(VBA\)](#)

Remarks

Before calling this method, you must have selected the reference entities using these marks with [IModelDocExtension::SelectByID2](#):

- 0 = First reference entity
- 1 = Second reference entity
- 2 = Third reference entity

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IRefPlane Interface](#)
[IRefPlaneFeatureData Interface](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertRevolvedRefSurface Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertRevolvedRefSurface Method (IFeatureManager)

Creates a revolved reference surface by revolving a profile around a centerline.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertRevolvedRefSurface( _  
    ByVal Angle As System.Double, _  
    ByVal ReverseDir As System.Boolean, _  
    ByVal Angle2 As System.Double, _  
    ByVal RevType As System.Integer _  
    ) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Angle As System.Double  
Dim ReverseDir As System.Boolean  
Dim Angle2 As System.Double  
Dim RevType As System.Integer  
Dim value As Feature
```

```
value = instance.InsertRevolvedRefSurface(Angle, ReverseDir, Angle2, RevType)
```

C#

```
Feature InsertRevolvedRefSurface(  
    System.double Angle,  
    System.bool ReverseDir,  
    System.double Angle2,  
    System.int RevType  
)
```

C++/CLI

```
Feature^ InsertRevolvedRefSurface(  
& System.double Angle,  
& System.bool ReverseDir,  
& System.double Angle2,  
& System.int RevType  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Angle

Angle of revolution in radians

ReverseDir

Angle is positive or negative (True or false)

Angle2

Angle of revolution in radians

RevType

Type of revolution (see **Remarks**)

Return Value

Pointer to [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertRevolvedRefSurface](#).

Remarks

Make the selections using [IModelDocExtension::SelectByID2](#) before calling this method. See the SOLIDWORKS Help for information about what entities are valid for selection.

The RevType argument can be one of these values:

- 0 = One direction revolution.
- 1 = MidPlane revolution. For this type of revolve, the angle specification specifies the full revolution. The angle to revolve is (angle/2) on either side of the sketch. The ReverseDir argument has no effect.
- 2 = Two direction revolution. For a two direction revolve, Angle is the angle to revolve in Direction1 and Angle2 is the angle to revolve in Direction2.

This method does not support 3D sketches.

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

[IRvolveFeatureData2 Interface](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertRib Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertRib Method (IFeatureManager)

Inserts a rib.

.NET Syntax

```

Sub InsertRib( _
    ByVal Is2Sided As System.Boolean, _
    ByVal ReverseThicknessDir As System.Boolean, _
    ByVal Thickness As System.Double, _
    ByVal ReferenceEdgeIndex As System.Integer, _
    ByVal ReverseMaterialDir As System.Boolean, _
    ByVal IsDrafted As System.Boolean, _
    ByVal DraftOutward As System.Boolean, _
    ByVal DraftAngle As System.Double, _
    ByVal IsNormToSketch As System.Boolean, _
    ByVal IsDraftedFromWall As System.Boolean _
)

```



```

Dim instance As IFeatureManager
Dim Is2Sided As System.Boolean
Dim ReverseThicknessDir As System.Boolean
Dim Thickness As System.Double
Dim ReferenceEdgeIndex As System.Integer
Dim ReverseMaterialDir As System.Boolean
Dim IsDrafted As System.Boolean
Dim DraftOutward As System.Boolean
Dim DraftAngle As System.Double
Dim IsNormToSketch As System.Boolean
Dim IsDraftedFromWall As System.Boolean

instance.InsertRib(Is2Sided, ReverseThicknessDir, Thickness, ReferenceEdgeIndex,
ReverseMaterialDir, IsDrafted, DraftOutward, DraftAngle, IsNormToSketch, IsDraftedFromWall)

```

```

void InsertRib(
    System.bool Is2Sided,
    System.bool ReverseThicknessDir,
    System.double Thickness,
    System.int ReferenceEdgeIndex,
    System.bool ReverseMaterialDir,
    System.bool IsDrafted,
    System.bool DraftOutward,
    System.double DraftAngle,
    System.bool IsNormToSketch,
    System.bool IsDraftedFromWall
)

```

```

void InsertRib(
    & System.bool Is2Sided,
    & System.bool ReverseThicknessDir,
    & System.double Thickness,
    & System.int ReferenceEdgeIndex,
    & System.bool ReverseMaterialDir,
    & System.bool IsDrafted,
    & System.bool DraftOutward,
    & System.double DraftAngle,
    & System.bool IsNormToSketch,
    & System.bool IsDraftedFromWall
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Is2Sided

True if the rib is double-sided, false if the rib is single sided; a double-sided rib thickens on both sides of the sketch

ReverseThicknessDir

True to thicken the opposite side of the sketch normal, false to not; only valid for single-sided ribs

Thickness

Rib thickness

ReferenceEdgeIndex

Edge in the sketch to use to determine the material direction and for draft reference; when the rib is drafted, the mid point of this edge maintains the thickness value; other points on the rib may have a different thicknesses due to the draft

ReverseMaterialDir

True to flip the direction of the extrusion, false to not

IsDrafted

True to draft the rib, false to not

DraftOutward

True to draft the rib outward, false to not; only valid if IsDrafted is true

DraftAngle

Draft angle of the rib; only valid if IsDrafted is true

IsNormToSketch

True if extrusion direction is normal to sketch, false if parallel to sketch

IsDraftedFromWall

True if draft is from wall, false if not; only valid if IsDrafted is true

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertRib](#).

Example

[Get Rib Feature Data \(VBA\)](#)

[Get Rib Feature Data \(VB.NET\)](#)

[Get Rib Feature Data \(C#\)](#)

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

[IRibFeatureData2 Interface](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertRuledSurfaceFromEdge Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertRuledSurfaceFromEdge Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::InsertRuledSurfaceFromEdge2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertRuledSurfaceFromEdge( _  
    ByVal Type As System.Integer, _  
    ByVal Length As System.Double, _  
    ByVal FlipPullDir As System.Boolean, _  
    ByVal FlipDir As System.Boolean, _  
    ByVal TrimAndSew As System.Boolean, _  
    ByVal Angle As System.Double, _  
    ByVal CoordInput As System.Boolean, _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
) As Feature
```

Visual Basic (Usage)

```
Dim instance As IFeatureManager  
Dim Type As System.Integer  
Dim Length As System.Double  
Dim FlipPullDir As System.Boolean  
Dim FlipDir As System.Boolean  
Dim TrimAndSew As System.Boolean  
Dim Angle As System.Double  
Dim CoordInput As System.Boolean
```



```
Dim X As System.Double
Dim Y As System.Double
Dim Z As System.Double
Dim value As Feature
```

```
value = instance.InsertRuledSurfaceFromEdge(Type, Length, FlipPullDir, FlipDir, TrimAndSew, Angle, CoordInput, X, Y, Z)
```

C#

```
Feature InsertRuledSurfaceFromEdge(
    System.int Type,
    System.double Length,
    System.bool FlipPullDir,
    System.bool FlipDir,
    System.bool TrimAndSew,
    System.double Angle,
    System.bool CoordInput,
    System.double X,
    System.double Y,
    System.double Z
)
```

C++/CLI

```
Feature^ InsertRuledSurfaceFromEdge(
    & System.int Type,
    & System.double Length,
    & System.bool FlipPullDir,
    & System.bool FlipDir,
    & System.bool TrimAndSew,
    & System.double Angle,
    & System.bool CoordInput,
    & System.double X,
    & System.double Y,
    & System.double Z
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Type

Length

FlipPullDir

FlipDir

TrimAndSew

Angle

CoordInput

X

Y

Z

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertRuledSurfaceFromEdge](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Inserts a ruled surface from the selected edge on this feature.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertRuledSurfaceFromEdge2( _
    ByVal Type As System.Integer, _
    ByVal Length As System.Double, _
    ByVal FlipPullDir As System.Boolean, _
    ByVal FlipDir As System.Boolean, _
    ByVal TrimAndSew As System.Boolean, _
    ByVal Angle As System.Double, _
    ByVal CoordInput As System.Boolean, _
    ByVal X As System.Double, _
    ByVal Y As System.Double, _
    ByVal Z As System.Double, _
    ByVal BConnectSurface As System.Boolean _
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager
Dim Type As System.Integer
Dim Length As System.Double
Dim FlipPullDir As System.Boolean
Dim FlipDir As System.Boolean
Dim TrimAndSew As System.Boolean
Dim Angle As System.Double
Dim CoordInput As System.Boolean
Dim X As System.Double
Dim Y As System.Double
Dim Z As System.Double
Dim BConnectSurface As System.Boolean
Dim value As Feature
```

```
value = instance.InsertRuledSurfaceFromEdge2(Type, Length, FlipPullDir, FlipDir, TrimAndSew, Angle, CoordInput, X, Y, Z, BConnectSurface)
```

C#

```
Feature InsertRuledSurfaceFromEdge2(
    System.int Type,
    System.double Length,
    System.bool FlipPullDir,
    System.bool FlipDir,
    System.bool TrimAndSew,
    System.double Angle,
    System.bool CoordInput,
    System.double X,
    System.double Y,
    System.double Z,
    System.bool BConnectSurface
)
```

C++/CLI

```
Feature^ InsertRuledSurfaceFromEdge2(
    & System.int Type,
    & System.double Length,
    & System.bool FlipPullDir,
    & System.bool FlipDir,
    & System.bool TrimAndSew,
    & System.double Angle,
    & System.bool CoordInput,
    & System.double X,
    & System.double Y,
    & System.double Z,
```

```
& System.bool BConnectSurface  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Type

- 0 = Tangent to Surface
- 1 = Normal to Surface
- 2 = Tapered to Vector
- 3 = Perpendicular to Vector
- 4 = Sweep

Length

Distance at which to create the surface; valid for Tangent to Surface, Tapered to Vector, Perpendicular to Vector, and Sweep types only

FlipPullDir

True to flip the pull direction, false to not; valid for Normal to Surface and Tapered to Vector types only

FlipDir

True to flip the direction, false to not; valid for Perpendicular to Vector type only

TrimAndSew

True to trim and knit the surface, false to not

Angle

Angle for Tapered to Vector type only

CoordInput

True to enable coordinate input, false if not; for Sweep type only

X

x coordinate

Y

y coordinate

Z

z coordinate

BConnectSurface

True to remove any connecting surfaces, false to not

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertRuledSurfaceFromEdge2](#).

Remarks

If you [select](#) an edge with a selection mark of 4, the default face is used. If you select an edge with a selection mark of 6, the alternate face is used.

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

[IRuledSurfaceFeatureData Interface](#)

[IModeler::CreateRuledSurface Method](#)

[IModeler::CreateRuledSurfaceFromEdge Method](#)

[IModeler::ICreateRuledSurface Method](#)

[IModeler::ICreateRuledSurfaceFromEdge Method](#)

Availability

SOLIDWORKS 2004 SP3, Revision Number 12.3

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertSaveOutBodies Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertSaveOutBodies Method (IFeatureManager)

Saves the selected surface bodies or solid bodies or sub-weldments to a file.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertSaveOutBodies() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim value As System.Boolean  
  
value = instance.InsertSaveOutBodies()
```

C#

```
System.bool InsertSaveOutBodies()
```

C++/CLI

```
System.bool InsertSaveOutBodies();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the selected surface bodies or solid bodies or sub-weldments are saved to a file, false if not

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertSaveOutBodies](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertScale Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertScale Method (IFeatureManager)

Applies the specified scaling to either the current model or a selected graphic body.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertScale( _  
    ByVal Type As System.Short, _  
    ByVal Uniform As System.Boolean, _  
    ByVal Xscale As System.Double, _  
    ByVal YScale As System.Double, _  
    ByVal ZScale As System.Double _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Type As System.Short  
Dim Uniform As System.Boolean  
Dim Xscale As System.Double  
Dim YScale As System.Double  
Dim ZScale As System.Double
```

Dim value As Feature

```
value = instance.InsertScale(Type, Uniform, Xscale, YScale, ZScale)
```

C#

```
Feature InsertScale(  
    System.short Type,  
    System.bool Uniform,  
    System.double Xscale,  
    System.double YScale,  
    System.double ZScale  
)
```

C++/CLI

```
Feature^ InsertScale(  
& System.short Type,  
& System.bool Uniform,  
& System.double Xscale,  
& System.double YScale,  
& System.double ZScale  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Type

Value as defined in [swScaleType_e](#)

Uniform

True if scaling should be uniform, false if not

Xscale

X direction scale factor

YScale

Y direction scale factor; valid only if Uniform is false

ZScale

Z direction scale factor; valid only if Uniform is false

Return Value

[IFeature](#) if scaling the current model; Null or Nothing if scaling a selected graphic body

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertScale](#).

Example

[Insert Scale Feature \(VBA\)](#)

Remarks

To scale a graphic body:

1. Select the graphic body in the Graphic Bodies folder of the FeatureManager design tree.
2. Call this method.

This method scales the selected graphic body up in the graphics view, but no scale feature is created.

If Uniform is...

True

False

Then the scaling factor is...

Xscale, applied in three directions.

All three scaling factors (Xscale, YScale, and ZScale).

Use [IFeature::GetDefinition](#) and [IFeature::IGetDefinition](#) and [IFeature::ModifyDefinition](#) and [IFeature::IModifyDefinition2](#) to gain access to the [IScaleFeatureData](#) object.

See Also

Availability

SOLIDWORKS 2003 SP2, Revision Number 11.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertSecurityNote Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertSecurityNote Method (IFeatureManager)

Inserts a note for the specified [macro feature](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertSecurityNote( _  
    ByVal Text As System.String, _  
    ByVal FeatureOwner As Feature _  
) As Note
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Text As System.String  
Dim FeatureOwner As Feature  
Dim value As Note  
  
value = instance.InsertSecurityNote(Text, FeatureOwner)
```

C#

```
Note InsertSecurityNote(  
    System.string Text,  
    Feature FeatureOwner  
)
```

C++/CLI

```
Note^ InsertSecurityNote(  
& System.String^ Text,  
& Feature^ FeatureOwner  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Text
Text for note
FeatureOwner
Macro feature for this note

Return Value

Point to [INote](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertSecurityNote](#).

Remarks

The note is for display purposes only and cannot be modified by the end user. For example, you could display the note to inform an end user the editing or deleting the macro feature is prohibited.

You associate the note with the specified macro feature in:

- VB by using this method in the code that generates the macro feature and by setting swMacroFeatureSecurityEnableNote in the [macro feature's security function](#). You should also include the conditions under which to display the note.

- C++ by using this method in the code that generates the macro feature and by setting swMacroFeatureSecurityEnableNote for the Options argument of [ISwComFeature::Security](#) in the [macro feature's security function](#). You should also include the conditions under which to display the note.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertSewRefSurface Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertSewRefSurface Method (IFeatureManager)

Creates a surface by knitting the selected surfaces together.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertSewRefSurface( _
    ByVal UseGapFilters As System.Boolean, _
    ByVal TryToFormSolid As System.Boolean, _
    ByVal MergeEntities As System.Boolean, _
    ByVal KnitTolerance As System.Double, _
    ByVal MaxValueForGapRange As System.Double _
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager
Dim UseGapFilters As System.Boolean
Dim TryToFormSolid As System.Boolean
Dim MergeEntities As System.Boolean
Dim KnitTolerance As System.Double
Dim MaxValueForGapRange As System.Double
Dim value As Feature
```

```
value = instance.InsertSewRefSurface(UseGapFilters, TryToFormSolid, MergeEntities, KnitTolerance,
MaxValueForGapRange)
```

C#

```
Feature InsertSewRefSurface(
    System.bool UseGapFilters,
    System.bool TryToFormSolid,
    System.bool MergeEntities,
    System.double KnitTolerance,
    System.double MaxValueForGapRange
)
```

C++/CLI

```
Feature^ InsertSewRefSurface(
    & System.bool UseGapFilters,
    & System.bool TryToFormSolid,
    & System.bool MergeEntities,
    & System.double KnitTolerance,
    & System.double MaxValueForGapRange
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UseGapFilters
True to use gap filters, false to not
TryToFormSolid
True to form a solid body, false to no
MergeEntities
True to merge edges and faces by removing redundant vertices and edges, false to not
KnitTolerance
Knit tolerance (see **Remarks**)
MaxValueForGapRange
Maximum value of gap range for gap filters

Return Value

[Feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertSewRefSurface](#).

Example

[Create Surface Knit Feature \(VB.NET\)](#)
[Create Surface Knit Feature \(VBA\)](#)
[Create Surface Knit Feature \(C#\)](#)

Remarks

Make selections using [IModelDocExtension::SelectByID2](#) with a mark number of 1. See the SOLIDWORKS Help for information about what entities are valid for selection.

A knit tolerance's:

- lower limit is 0.0001 mm
- upper limit is 0.1 mm

The knit tolerance value should be in a gap range such that:

(Minimum gap) <= (Knit tolerance) <= MIN(0.1 mm, Maximum gap)

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[ISurfaceKnitFeatureData Interface](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertSheetMetal3dBend Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertSheetMetal3dBend Method (IFeatureManager)

Inserts a 3D bend in sheet metal part.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertSheetMetal3dBend( _  
    ByVal Angle As System.Double, _  
    ByVal BUseDefaultRadius As System.Boolean, _  
    ByVal Radius As System.Double, _  
    ByVal FlipDir As System.Boolean, _  
    ByVal BendPos As System.Short, _  
    ByVal PCBA As CustomBendAllowance _  
) As Feature
```

Visual Basic (Usage)

```
Dim instance As IFeatureManager  
Dim Angle As System.Double  
Dim BUseDefaultRadius As System.Boolean
```



```

Dim Radius As System.Double
Dim FlipDir As System.Boolean
Dim BendPos As System.Short
Dim PCBA As CustomBendAllowance
Dim value As Feature

value = instance.InsertSheetMetal3dBend(Angle, BUseDefaultRadius, Radius, FlipDir, BendPos, PCBA)

```

C#

```

Feature InsertSheetMetal3dBend(
    System.double Angle,
    System.bool BUseDefaultRadius,
    System.double Radius,
    System.bool FlipDir,
    System.short BendPos,
    CustomBendAllowance PCBA
)

```

C++/CLI

```

Feature^ InsertSheetMetal3dBend(
    & System.double Angle,
    & System.bool BUseDefaultRadius,
    & System.double Radius,
    & System.bool FlipDir,
    & System.short BendPos,
    & CustomBendAllowance^ PCBA
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Angle

Angle of the bend in radians

BUseDefaultRadius

True to use the default radius, false to use the value specified in radius

Radius

Value for the radius of the bend if bUseDefaultRadius is false

FlipDir

True to flip the bend direction, false to not

BendPos

Bend position:

- 0 = bend centerline
- 1 = material inside
- 2 = material outside
- 3 = bend outside

PCBA

If...	Then...
non-NULL	Pointer to ICustomBendAllowance object for which required values have been set
NULL	Parent bend's bend allowance is used

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertSheetMetal3dBend](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[ISheetMetalFeatureData Interface](#)
[IBendsFeatureData Interface](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertSheetMetalBaseFlange Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertSheetMetalBaseFlange Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager.InsertSheetMetalBaseFlange2](#)

.NET Syntax

Visual Basic (Declaration)

```
Function InsertSheetMetalBaseFlange( _  
    ByVal Thickness As System.Double, _  
    ByVal ThickenDir As System.Boolean, _  
    ByVal Radius As System.Double, _  
    ByVal ExtrudeDist1 As System.Double, _  
    ByVal ExtrudeDist2 As System.Double, _  
    ByVal FlipExtruDir As System.Boolean, _  
    ByVal EndCondition1 As System.Integer, _  
    ByVal EndCondition2 As System.Integer, _  
    ByVal DirToUse As System.Integer, _  
    ByVal PCBA As CustomBendAllowance, _  
    ByVal UseDefaultRelief As System.Boolean, _  
    ByVal ReliefType As System.Integer, _  
    ByVal ReliefWidth As System.Double, _  
    ByVal ReliefDepth As System.Double, _  
    ByVal ReliefRatio As System.Double, _  
    ByVal UseReliefRatio As System.Boolean _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Thickness As System.Double  
Dim ThickenDir As System.Boolean  
Dim Radius As System.Double  
Dim ExtrudeDist1 As System.Double  
Dim ExtrudeDist2 As System.Double  
Dim FlipExtruDir As System.Boolean  
Dim EndCondition1 As System.Integer  
Dim EndCondition2 As System.Integer  
Dim DirToUse As System.Integer  
Dim PCBA As CustomBendAllowance  
Dim UseDefaultRelief As System.Boolean  
Dim ReliefType As System.Integer  
Dim ReliefWidth As System.Double  
Dim ReliefDepth As System.Double  
Dim ReliefRatio As System.Double  
Dim UseReliefRatio As System.Boolean  
Dim value As Feature
```

```
value = instance.InsertSheetMetalBaseFlange(Thickness, ThickenDir, Radius, ExtrudeDist1,  
ExtrudeDist2, FlipExtruDir, EndCondition1, EndCondition2, DirToUse, PCBA, UseDefaultRelief,  
ReliefType, ReliefWidth, ReliefDepth, ReliefRatio, UseReliefRatio)
```

C#

```
Feature InsertSheetMetalBaseFlange(  
    System.double Thickness,  
    System.bool ThickenDir,  
    System.double Radius,  
    System.double ExtrudeDist1,  
    System.double ExtrudeDist2,  
    System.bool FlipExtruDir,
```

```

System.int EndCondition1,
System.int EndCondition2,
System.int DirToUse,
CustomBendAllowance PCBA,
System.bool UseDefaultRelief,
System.int ReliefType,
System.double ReliefWidth,
System.double ReliefDepth,
System.double ReliefRatio,
System.bool UseReliefRatio
)

```

C++/CLI

```

Feature^ InsertSheetMetalBaseFlange(
& System.double Thickness,
& System.bool ThickenDir,
& System.double Radius,
& System.double ExtrudeDist1,
& System.double ExtrudeDist2,
& System.bool FlipExtruDir,
& System.int EndCondition1,
& System.int EndCondition2,
& System.int DirToUse,
& CustomBendAllowance^ PCBA,
& System.bool UseDefaultRelief,
& System.int ReliefType,
& System.double ReliefWidth,
& System.double ReliefDepth,
& System.double ReliefRatio,
& System.bool UseReliefRatio
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Thickness

Wall thickness of the sheet metal feature

ThickenDir

Direction to thicken the sheet metal

Radius

Global bend radius to insert at the corners

ExtrudeDist1

Distance of the sheet metal extrusion for the Direction 1

ExtrudeDist2

Distance of the sheet metal extrusion for the Direction 2

FlipExtruDir

True to reverse the direction of the extrude, false to not

EndCondition1

End condition for first extrude distance

EndCondition2

End condition for second extrude distance

DirToUse

End condition type as defined in [swEndConditions_e](#)

PCBA

Pointer to the [custom bend allowance](#) object

UseDefaultRelief

True to use the default relief, false to not

ReliefType

Relief type as defined in [swSheetMetalReliefTypes_e](#)

ReliefWidth

Width of the relief

ReliefDepth

Depth of the relief

ReliefRatio

Value for the relief ratio

UseReliefRatio

True to use the relief ratio, false to not

Return Value

Pointer to the [IFeature](#) object

See [FeatureManager::InsertSheetMetalBaseFlange](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IBaseFlangeFeatureData Interface](#)
[ISheetMetalFeatureData Interface](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertSheetMetalBaseFlange2 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertSheetMetalBaseFlange2 Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::CreateDefinition](#) and [IFeatureManager::CreateFeature](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertSheetMetalBaseFlange2( _  
    ByVal Thickness As System.Double, _  
    ByVal ThickenDir As System.Boolean, _  
    ByVal Radius As System.Double, _  
    ByVal ExtrudeDist1 As System.Double, _  
    ByVal ExtrudeDist2 As System.Double, _  
    ByVal FlipExtruDir As System.Boolean, _  
    ByVal EndCondition1 As System.Integer, _  
    ByVal EndCondition2 As System.Integer, _  
    ByVal DirToUse As System.Integer, _  
    ByVal PCBA As CustomBendAllowance, _  
    ByVal UseDefaultRelief As System.Boolean, _  
    ByVal ReliefType As System.Integer, _  
    ByVal ReliefWidth As System.Double, _  
    ByVal ReliefDepth As System.Double, _  
    ByVal ReliefRatio As System.Double, _  
    ByVal UseReliefRatio As System.Boolean, _  
    ByVal Merge As System.Boolean, _  
    ByVal UseFeatScope As System.Boolean, _  
    ByVal UseAutoSelect As System.Boolean _  
) As Feature
```

Visual Basic (Usage)

```
Dim instance As IFeatureManager  
Dim Thickness As System.Double  
Dim ThickenDir As System.Boolean  
Dim Radius As System.Double  
Dim ExtrudeDist1 As System.Double  
Dim ExtrudeDist2 As System.Double  
Dim FlipExtruDir As System.Boolean  
Dim EndCondition1 As System.Integer  
Dim EndCondition2 As System.Integer  
Dim DirToUse As System.Integer  
Dim PCBA As CustomBendAllowance  
Dim UseDefaultRelief As System.Boolean  
Dim ReliefType As System.Integer  
Dim ReliefWidth As System.Double  
Dim ReliefDepth As System.Double  
Dim ReliefRatio As System.Double  
Dim UseReliefRatio As System.Boolean  
Dim Merge As System.Boolean  
Dim UseFeatScope As System.Boolean  
Dim UseAutoSelect As System.Boolean
```



Dim value As Feature

```
value = instance.InsertSheetMetalBaseFlange2(Thickness, ThickenDir, Radius, ExtrudeDist1, ExtrudeDist2, FlipExtruDir, EndCondition1, EndCondition2, DirToUse, PCBA, UseDefaultRelief, ReliefType, ReliefWidth, ReliefDepth, ReliefRatio, UseReliefRatio, Merge, UseFeatScope, UseAutoSelect)
```

C#

```
Feature InsertSheetMetalBaseFlange2(  
    System.double Thickness,  
    System.bool ThickenDir,  
    System.double Radius,  
    System.double ExtrudeDist1,  
    System.double ExtrudeDist2,  
    System.bool FlipExtruDir,  
    System.int EndCondition1,  
    System.int EndCondition2,  
    System.int DirToUse,  
    CustomBendAllowance PCBA,  
    System.bool UseDefaultRelief,  
    System.int ReliefType,  
    System.double ReliefWidth,  
    System.double ReliefDepth,  
    System.double ReliefRatio,  
    System.bool UseReliefRatio,  
    System.bool Merge,  
    System.bool UseFeatScope,  
    System.bool UseAutoSelect  
)
```

C++/CLI

```
Feature^ InsertSheetMetalBaseFlange2(  
& System.double Thickness,  
& System.bool ThickenDir,  
& System.double Radius,  
& System.double ExtrudeDist1,  
& System.double ExtrudeDist2,  
& System.bool FlipExtruDir,  
& System.int EndCondition1,  
& System.int EndCondition2,  
& System.int DirToUse,  
& CustomBendAllowance^ PCBA,  
& System.bool UseDefaultRelief,  
& System.int ReliefType,  
& System.double ReliefWidth,  
& System.double ReliefDepth,  
& System.double ReliefRatio,  
& System.bool UseReliefRatio,  
& System.bool Merge,  
& System.bool UseFeatScope,  
& System.bool UseAutoSelect  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Thickness

Wall thickness of the sheet metal feature

ThickenDir

Direction to thicken the sheet metal

Radius

Global bend radius to insert at the corners

ExtrudeDist1

Distance of the sheet metal extrusion for the Direction 1

ExtrudeDist2

Distance of the sheet metal extrusion for the Direction 2

FlipExtruDir

True to reverse the direction of the extrude, false to not

EndCondition1

End condition for first extrude distance
EndCondition2
End condition for second extrude distance
DirToUse
End condition type as defined in [swEndConditions_e](#)
PCBA
Pointer to the [custom bend allowance](#) object
UseDefaultRelief
True to use the default relief, false to not
ReliefType
Relief type as defined in [swSheetMetalReliefTypes_e](#)
ReliefWidth
Width of the relief
ReliefDepth
Depth of the relief
ReliefRatio
Value for the relief ratio
UseReliefRatio
True to use the relief ratio, false to not
Merge
True to connect bodies into a single sheet metal body; false to not
UseFeatScope
True to use selected bodies, false to use all bodies
UseAutoSelect
True to auto select bodies (see **Remarks**)

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertSheetMetalBaseFlange2](#).

Example

[Insert Sheet Metal Base Flange \(VBA\)](#)
[Insert Sheet Metal Base Flange \(VB.NET\)](#)
[Insert Sheet Metal Base Flange \(C#\)](#)

Remarks

When UseAutoSelect is set to false and bodies are selected using [IModelDocExtension::SelectById2](#) or other select methods, use selection mark 8 for the bodies.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IBaseFlangeFeatureData Interface](#)
[ISheetMetalFeatureData Interface](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertSheetMetalCornerTrim Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertSheetMetalCornerTrim Method (IFeatureManager)

Inserts a break corner trim in the sheet metal part.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertSheetMetalCornerTrim(  
    ByVal InternalCornerFlag As System.Integer, _  
    ByVal BreakType As System.Integer, _  
    ByVal BreakDist As System.Double, _  
    ByVal ReliefType As System.Integer, _  
    ByVal Param As System.Double _  
) As Feature
```

```

Dim instance As IFeatureManager
Dim InternalCornerFlag As System.Integer
Dim BreakType As System.Integer
Dim BreakDist As System.Double
Dim ReliefType As System.Integer
Dim Param As System.Double
Dim value As Feature

value = instance.InsertSheetMetalCornerTrim(InternalCornerFlag, BreakType, BreakDist, ReliefType, Param)

```

C#

```

Feature InsertSheetMetalCornerTrim(
    System.int InternalCornerFlag,
    System.int BreakType,
    System.double BreakDist,
    System.int ReliefType,
    System.double Param
)

```

C++/CLI

```

Feature^ InsertSheetMetalCornerTrim(
    & System.int InternalCornerFlag,
    & System.int BreakType,
    & System.double BreakDist,
    & System.int ReliefType,
    & System.double Param
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

InternalCornerFlag

Do internal corners only

BreakType

Type of break as defined in [swBreakCornerTypes_e](#)

BreakDist

Distance to break from corner

ReliefType

Type of relief:

- 0 = circular
- 1 = square
- 2 = bend-waist

Param

ReliefType dependent:

- circular, its diameter
- square, its side length
- bend-waist, its radius

Return Value

Pointer to the break corner trim [feature](#) in the sheet metal part

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertSheetMetalCornerTrim](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IBreakCornerFeatureData Interface](#)
[ISheetMetalFeatureData Interface](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertSheetMetalEdgeFlange Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertSheetMetalEdgeFlange Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager.InsertSheetMetalEdgeFlange2](#)

.NET Syntax

Visual Basic (Declaration)

```
Function InsertSheetMetalEdgeFlange( _  
    ByVal FlangeEdge As Edge, _  
    ByVal SketchFeat As Feature, _  
    ByVal BooleanOptions As System.Integer, _  
    ByVal DAngle As System.Double, _  
    ByVal DRadius As System.Double, _  
    ByVal BendPosition As System.Integer, _  
    ByVal DOffsetDist As System.Double, _  
    ByVal ReliefType As System.Integer, _  
    ByVal DReliefRatio As System.Double, _  
    ByVal DReliefWidth As System.Double, _  
    ByVal DReliefDepth As System.Double, _  
    ByVal FlangeSharpType As System.Integer, _  
    ByVal PCBA As CustomBendAllowance _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim FlangeEdge As Edge  
Dim SketchFeat As Feature  
Dim BooleanOptions As System.Integer  
Dim DAngle As System.Double  
Dim DRadius As System.Double  
Dim BendPosition As System.Integer  
Dim DOffsetDist As System.Double  
Dim ReliefType As System.Integer  
Dim DReliefRatio As System.Double  
Dim DReliefWidth As System.Double  
Dim DReliefDepth As System.Double  
Dim FlangeSharpType As System.Integer  
Dim PCBA As CustomBendAllowance  
Dim value As Feature
```

```
value = instance.InsertSheetMetalEdgeFlange(FlangeEdge, SketchFeat, BooleanOptions, DAngle,  
DRadius, BendPosition, DOffsetDist, ReliefType, DReliefRatio, DReliefWidth, DReliefDepth,  
FlangeSharpType, PCBA)
```

C#

```
Feature InsertSheetMetalEdgeFlange(  
    Edge FlangeEdge,  
    Feature SketchFeat,  
    System.int BooleanOptions,  
    System.double DAngle,  
    System.double DRadius,  
    System.int BendPosition,  
    System.double DOffsetDist,  
    System.int ReliefType,  
    System.double DReliefRatio,  
    System.double DReliefWidth,  
    System.double DReliefDepth,  
    System.int FlangeSharpType,
```

CustomBendAllowance [PCBA](#)

)

C++/CLI

```
Feature^ InsertSheetMetalEdgeFlange(  
& Edge^ FlangeEdge,  
& Feature^ SketchFeat,  
& System.int BooleanOptions,  
& System.double DAngle,  
& System.double DRadius,  
& System.int BendPosition,  
& System.double DOffsetDist,  
& System.int ReliefType,  
& System.double DReliefRatio,  
& System.double DReliefWidth,  
& System.double DReliefDepth,  
& System.int FlangeSharpType,  
& CustomBendAllowance^ PCBA  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FlangeEdge

SketchFeat

BooleanOptions

DAngle

DRadius

BendPosition

DOffsetDist

ReliefType

DReliefRatio

DReliefWidth

DReliefDepth

FlangeSharpType

PCBA

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertSheetMetalEdgeFlange](#).

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertSheetMetalEdgeFlange2 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertSheetMetalEdgeFlange2 Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::CreateDefinition](#) and [IFeatureManager::CreateFeature](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertSheetMetalEdgeFlange2( _  
    ByVal FlangeEdges As System.Object, _  
    ByVal SketchFeats As System.Object, _  
    ByVal BooleanOptions As System.Integer, _  
    ByVal FlangeAngle As System.Double, _  
    ByVal FlangeRadius As System.Double, _
```

```

    ByVal BendPosition As System.Integer, _
    ByVal FlangeOffsetDist As System.Double, _
    ByVal ReliefType As System.Integer, _
    ByVal FlangeReliefRatio As System.Double, _
    ByVal FlangeReliefWidth As System.Double, _
    ByVal FlangeReliefDepth As System.Double, _
    ByVal FlangeSharpType As System.Integer, _
    ByVal CustomBendAllowance As CustomBendAllowance _
) As Feature

```

Visual Basic (Usage)



```

Dim instance As IFeatureManager
Dim FlangeEdges As System.Object
Dim SketchFeats As System.Object
Dim BooleanOptions As System.Integer
Dim FlangeAngle As System.Double
Dim FlangeRadius As System.Double
Dim BendPosition As System.Integer
Dim FlangeOffsetDist As System.Double
Dim ReliefType As System.Integer
Dim FlangeReliefRatio As System.Double
Dim FlangeReliefWidth As System.Double
Dim FlangeReliefDepth As System.Double
Dim FlangeSharpType As System.Integer
Dim CustomBendAllowance As CustomBendAllowance
Dim value As Feature

```

```

value = instance.InsertSheetMetalEdgeFlange2(FlangeEdges, SketchFeats, BooleanOptions,
FlangeAngle, FlangeRadius, BendPosition, FlangeOffsetDist, ReliefType, FlangeReliefRatio,
FlangeReliefWidth, FlangeReliefDepth, FlangeSharpType, CustomBendAllowance)

```

C#

```

Feature InsertSheetMetalEdgeFlange2(
    System.object FlangeEdges,
    System.object SketchFeats,
    System.int BooleanOptions,
    System.double FlangeAngle,
    System.double FlangeRadius,
    System.int BendPosition,
    System.double FlangeOffsetDist,
    System.int ReliefType,
    System.double FlangeReliefRatio,
    System.double FlangeReliefWidth,
    System.double FlangeReliefDepth,
    System.int FlangeSharpType,
    CustomBendAllowance CustomBendAllowance
)

```

C++/CLI

```

Feature^ InsertSheetMetalEdgeFlange2(
& System.Object^ FlangeEdges,
& System.Object^ SketchFeats,
& System.int BooleanOptions,
& System.double FlangeAngle,
& System.double FlangeRadius,
& System.int BendPosition,
& System.double FlangeOffsetDist,
& System.int ReliefType,
& System.double FlangeReliefRatio,
& System.double FlangeReliefWidth,
& System.double FlangeReliefDepth,
& System.int FlangeSharpType,
& CustomBendAllowance^ CustomBendAllowance
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FlangeEdges
Array of [edges](#) to which to apply a flange
SketchFeats
Array of [sketches](#) to use for the flange
BooleanOptions
Flange options as defined by [swInsertEdgeFlangeOptions_e](#)

FlangeAngle
Flange angle
FlangeRadius
Bend radius
BendPosition
Flange bend position as defined by [swFlangePositionTypes_e](#)
FlangeOffsetDist
Length of flange
ReliefType
Relief type as defined by [swSheetMetalReliefTypes_e](#)
FlangeReliefRatio
Relief ratio
FlangeReliefWidth
Relief width
FlangeReliefDepth
Relief depth
FlangeSharpType
Flange virtual sharp type as defined by [swFlangeDimTypes_e](#)
CustomBendAllowance

If...	Then...
non-NULL	Pointer to ICustomBendAllowance object for which required values have been set
NULL	Parent bend's bend allowance is used

Return Value

Pointer to the [IFeature](#) object

 Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertSheetMetalEdgeFlange2](#).

 Example

[Insert Sheet Metal Edge Flange \(VBA\)](#)
[Create Corner Relief Feature \(C#\)](#)
[Create Corner Relief Feature \(VBA\)](#)
[Create Corner Relief Feature \(VB.NET\)](#)

 Remarks

Before calling this method, call [IModelDoc2::InsertSketchForEdgeFlange](#) and create a profile for the flange. After creating the profile, call this method to create the flange.

 See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IEdgeFlangeFeatureData Interface](#)
[ISheetMetalFeatureData Interface](#)

 Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

InsertSheetMetalGussetFeature Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertSheetMetalGussetFeature Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::InsertSheetMetalGussetFeature3](#).

 .NET Syntax

Visual Basic (Declaration)

Function InsertSheetMetalGussetFeature(_
ByVal [BOffset](#) As System.Boolean, _

```

ByVal DOffset As System.Double, _
ByVal BFlipOffsetSide As System.Boolean, _
ByVal ProfDimType As System.Integer, _
ByVal DIndentDepth As System.Double, _
ByVal DLength As System.Double, _
ByVal BUseAngle As System.Boolean, _
ByVal DHeight As System.Double, _
ByVal DAngle As System.Double, _
ByVal BFlipSides As System.Boolean, _
ByVal DWidth As System.Double, _
ByVal DThickness As System.Double, _
ByVal BDraft As System.Boolean, _
ByVal DDraftAngle As System.Double, _
ByVal BInnerCornerFillet As System.Boolean, _
ByVal DInnerCornerFilletRadius As System.Double, _
ByVal BOuterCornerFillet As System.Boolean, _
ByVal DOuterCornerFilletRadius As System.Double, _
ByVal GussetType As System.Integer, _
ByVal BEdgeFillet As System.Boolean, _
ByVal DEdgeFilletRadius As System.Double _
) As Feature

```

Visual Basic (Usage)



```

Dim instance As IFeatureManager
Dim BOffset As System.Boolean
Dim DOffset As System.Double
Dim BFlipOffsetSide As System.Boolean
Dim ProfDimType As System.Integer
Dim DIndentDepth As System.Double
Dim DLength As System.Double
Dim BUseAngle As System.Boolean
Dim DHeight As System.Double
Dim DAngle As System.Double
Dim BFlipSides As System.Boolean
Dim DWidth As System.Double
Dim DThickness As System.Double
Dim BDraft As System.Boolean
Dim DDraftAngle As System.Double
Dim BInnerCornerFillet As System.Boolean
Dim DInnerCornerFilletRadius As System.Double
Dim BOuterCornerFillet As System.Boolean
Dim DOuterCornerFilletRadius As System.Double
Dim GussetType As System.Integer
Dim BEdgeFillet As System.Boolean
Dim DEdgeFilletRadius As System.Double
Dim value As Feature

```

```

value = instance.InsertSheetMetalGussetFeature(BOffset, DOffset, BFlipOffsetSide, ProfDimType,
DIndentDepth, DLength, BUseAngle, DHeight, DAngle, BFlipSides, DWidth, DThickness, BDraft,
DDraftAngle, BInnerCornerFillet, DInnerCornerFilletRadius, BOuterCornerFillet,
DOuterCornerFilletRadius, GussetType, BEdgeFillet, DEdgeFilletRadius)

```

C#

```

Feature InsertSheetMetalGussetFeature(
    System.bool BOffset,
    System.double DOffset,
    System.bool BFlipOffsetSide,
    System.int ProfDimType,
    System.double DIndentDepth,
    System.double DLength,
    System.bool BUseAngle,
    System.double DHeight,
    System.double DAngle,
    System.bool BFlipSides,
    System.double DWidth,
    System.double DThickness,
    System.bool BDraft,

```

```

    System.double DDraftAngle,
    System.bool BInnerCornerFillet,
    System.double DInnerCornerFilletRadius,
    System.bool BOuterCornerFillet,
    System.double DOuterCornerFilletRadius,
    System.int GussetType,
    System.bool BEdgeFillet,
    System.double DEdgeFilletRadius
)

```

C++/CLI

```

Feature^ InsertSheetMetalGussetFeature(
& System.bool BOffset,
& System.double DOffset,
& System.bool BFlipOffsetSide,
& System.int ProfDimType,
& System.double DIndentDepth,
& System.double DLength,
& System.bool BUseAngle,
& System.double DHeight,
& System.double DAngle,
& System.bool BFlipSides,
& System.double DWidth,
& System.double DThickness,
& System.bool BDraft,
& System.double DDraftAngle,
& System.bool BInnerCornerFillet,
& System.double DInnerCornerFilletRadius,
& System.bool BOuterCornerFillet,
& System.double DOuterCornerFilletRadius,
& System.int GussetType,
& System.bool BEdgeFillet,
& System.double DEdgeFilletRadius
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BOffset

True to offset the gusset section plane from the selected reference point, false to not

DOffset

Value by which to offset the gusset section plane from the selected reference point; valid only if BOffset is true

BFlipOffsetSide

True to specify the gusset section plane offset on the opposite side of the selected reference point, false to not; valid only if BOffset is true

ProfDimType

Type of gusset profile dimensioning scheme:

- 0 = indent
- 1 = profile dimensioning

(see **Remarks**)

DIndentDepth

Gusset depth; valid only if ProfDimType = 0

DLength

d1 gusset leg length; length of sheet metal from bend to intersection with first end of gusset; valid only if ProfDimType = 1 (see **Remarks**)

BUseAngle

True to dimension the angle that is formed by the intersection of the gusset with either face adjacent to the bend; valid only if ProfDimType = 1 (see **Remarks**)

DHeight

d2 gusset leg length; length of sheet metal from bend to intersection with second end of gusset; valid only if ProfDimType = 1 (see **Remarks**)

DAngle

Angle formed by intersection of gusset with one face adjacent to the bend; valid only if ProfDimType = 1 and BUseAngle is true (see **Remarks**)

BFlipSides

True to swap d1 and d2 in the dimensioning scheme, false to not; valid only if ProfDimType = 1 (see **Remarks**)

DWidth

Gusset width

DThickness

Gusset thickness

BDraft

True to draft gusset side faces, false to not

DDraftAngle

Angle by which to draft the gusset side faces; valid only if BDraft is true

BInnerCornerFillet

True to fillet the gusset inner corners, false to not

DInnerCornerFilletRadius

Inner corner fillet radius; valid only if BInnerCornerFillet is true

BOuterCornerFillet

True to fillet the gusset outer corners, false to not

DOuterCornerFilletRadius

Outer corner fillet radius; valid only if BOuterCornerFilletRadius is true

GussetType

Type of gusset:

- 0 = rounded back
- 1 = flat back

BEdgeFillet

True to fillet the edges of a gusset, false to not; valid only if GussetType = 1
DEdgeFilletRadius
Edge fillet radius; valid only if BEdgeFillet is true and GussetType = 1

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertSheetMetalGussetFeature](#).

Remarks

Before calling this method to create a sheet metal gusset feature, you must pre-select faces, lines, edges, vertices, or points as follows:

To...	Call IModelDocExtension::SelectByID2 to select...
Specify the legs of the gusset feature	- OR - - Two flat faces of a bend. - Two flat faces adjacent to a cylindrical bend face. - One cylindrical bend face. - One flat face and one non-planar face adjacent to a toroidal bend. - One toroidal bend face.
Optionally orient the gusset section plane	One linear edge or sketch segment that is perpendicular to the gusset section plane. If you do not pre-select either entity, then the method uses the bend line adjacent to the gusset legs as the reference.
Optionally position the gusset section plane	One vertex, sketch point, or reference point. If you do not pre-select any one of these entities, then you must set BOffset to true and specify DOffset for an assumed reference point. For insertion of the first gusset, the assumed reference point is on the end of the sheet metal bend. For second and subsequent gusset insertions, the assumed reference point is on the edge of the gusset last inserted.

Instead of pre-selecting the references and calling this method, you can pass the references as arrays to [IFeatureManager::InsertSheetMetalGussetFeature2](#).

If ProfDimType = 1, to dimension the gusset profile using...	Specify...
Both legs of the gusset	- DLength of d1 leg - DHeight of d2 leg - BUseAngle = false - BFlipSides = true to swap d1 and d2 legs during dimensioning
d1 leg and the angle it forms with the gusset	- DLength of d1 leg - DAngle of angle formed by gusset intersecting d1 leg - BUseAngle = true - BFlipSides = true to swap d1 and d2 legs during dimensioning
d2 leg and the angle it forms with the gusset	- DHeight of d2 leg - DAngle of angle formed by gusset intersecting d2 leg - BUseAngle = true - BFlipSides = true to swap d1 and d2 legs during dimensioning

See the SOLIDWORKS Help for more information about sheet metal gusset features.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[ISMGussetFeatureData Interface](#)

Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

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InsertSheetMetalGussetFeature2 Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertSheetMetalGussetFeature2 Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::InsertSheetMetalGussetFeature3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertSheetMetalGussetFeature2( _  
    ByVal BOffset As System.Boolean, _
```

```

ByVal DOffset As System.Double, _
ByVal BFlipOffsetSide As System.Boolean, _
ByVal ProfDimType As System.Integer, _
ByVal DIndentDepth As System.Double, _
ByVal DLength As System.Double, _
ByVal BUseAngle As System.Boolean, _
ByVal DHeight As System.Double, _
ByVal DAngle As System.Double, _
ByVal BFlipSides As System.Boolean, _
ByVal DWidth As System.Double, _
ByVal DThickness As System.Double, _
ByVal BDraft As System.Boolean, _
ByVal DDraftAngle As System.Double, _
ByVal BInnerCornerFillet As System.Boolean, _
ByVal DInnerCornerFilletRadius As System.Double, _
ByVal BOuterCornerFillet As System.Boolean, _
ByVal DOuterCornerFilletRadius As System.Double, _
ByVal GussetType As System.Integer, _
ByVal BEdgeFillet As System.Boolean, _
ByVal DEdgeFilletRadius As System.Double, _
ByVal ArrayOfFaces As System.Object, _
ByVal ArrayRefLines As System.Object, _
ByVal ArrayRefPoints As System.Object _
) As Feature

```

Visual Basic (Usage)



```

Dim instance As IFeatureManager
Dim BOffset As System.Boolean
Dim DOffset As System.Double
Dim BFlipOffsetSide As System.Boolean
Dim ProfDimType As System.Integer
Dim DIndentDepth As System.Double
Dim DLength As System.Double
Dim BUseAngle As System.Boolean
Dim DHeight As System.Double
Dim DAngle As System.Double
Dim BFlipSides As System.Boolean
Dim DWidth As System.Double
Dim DThickness As System.Double
Dim BDraft As System.Boolean
Dim DDraftAngle As System.Double
Dim BInnerCornerFillet As System.Boolean
Dim DInnerCornerFilletRadius As System.Double
Dim BOuterCornerFillet As System.Boolean
Dim DOuterCornerFilletRadius As System.Double
Dim GussetType As System.Integer
Dim BEdgeFillet As System.Boolean
Dim DEdgeFilletRadius As System.Double
Dim ArrayOfFaces As System.Object
Dim ArrayRefLines As System.Object
Dim ArrayRefPoints As System.Object
Dim value As Feature

```

```

value = instance.InsertSheetMetalGussetFeature2(BOffset, DOffset, BFlipOffsetSide, ProfDimType,
DIndentDepth, DLength, BUseAngle, DHeight, DAngle, BFlipSides, DWidth, DThickness, BDraft,
DDraftAngle, BInnerCornerFillet, DInnerCornerFilletRadius, BOuterCornerFillet,
DOuterCornerFilletRadius, GussetType, BEdgeFillet, DEdgeFilletRadius, ArrayOfFaces, ArrayRefLines,
ArrayRefPoints)

```

C#

```

Feature InsertSheetMetalGussetFeature2(
    System.bool BOffset,
    System.double DOffset,
    System.bool BFlipOffsetSide,
    System.int ProfDimType,
    System.double DIndentDepth,
    System.double DLength,

```

```

System.bool BUseAngle,
System.double DHeight,
System.double DAngle,
System.bool BFlipSides,
System.double DWidth,
System.double DThickness,
System.bool BDraft,
System.double DDraftAngle,
System.bool BInnerCornerFillet,
System.double DInnerCornerFilletRadius,
System.bool BOuterCornerFillet,
System.double DOuterCornerFilletRadius,
System.int GussetType,
System.bool BEdgeFillet,
System.double DEdgeFilletRadius,
System.object ArrayOfFaces,
System.object ArrayRefLines,
System.object ArrayRefPoints
)

```

C++/CLI

```

Feature^ InsertSheetMetalGussetFeature2(
& System.bool BOffset,
& System.double DOffset,
& System.bool BFlipOffsetSide,
& System.int ProfDimType,
& System.double DIndentDepth,
& System.double DLength,
& System.bool BUseAngle,
& System.double DHeight,
& System.double DAngle,
& System.bool BFlipSides,
& System.double DWidth,
& System.double DThickness,
& System.bool BDraft,
& System.double DDraftAngle,
& System.bool BInnerCornerFillet,
& System.double DInnerCornerFilletRadius,
& System.bool BOuterCornerFillet,
& System.double DOuterCornerFilletRadius,
& System.int GussetType,
& System.bool BEdgeFillet,
& System.double DEdgeFilletRadius,
& System.Object^ ArrayOfFaces,
& System.Object^ ArrayRefLines,
& System.Object^ ArrayRefPoints
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BOffset

True to offset the gusset section plane from the selected reference point in ArrayRefPoints, false to not

DOffset

Value by which to offset the gusset section plane from the selected reference point in ArrayRefPoints; valid only if BOffset is true

BFlipOffsetSide

True to specify the gusset section plane offset on the opposite side of the selected reference point, false to not; valid only if BOffset is true

ProfDimType

Type of gusset profile dimensioning scheme as defined in [swSheetMetalGussetProfileDimType_e](#)

(see **Remarks**)

DIndentDepth

Gusset indent depth; valid only if ProfDimType = 0

DLength

d1 gusset leg length; length of sheet metal from bend to intersection with first end of gusset; valid only if ProfDimType = 1 (see **Remarks**)

BUseAngle

True to dimension the angle that is formed by the intersection of the gusset with either face adjacent to the bend; valid only if ProfDimType = 1 (see **Remarks**)

DHeight

d2 gusset leg length; length of sheet metal from bend to intersection with second end of gusset; valid only if ProfDimType = 1 (see **Remarks**)

DAngle

Angle formed by intersection of gusset with one face adjacent to the bend; valid only if ProfDimType = 1 and BUseAngle is true (see **Remarks**)

BFlipSides

True to swap d1 and d2 legs in the dimensioning scheme, false to not; valid only if ProfDimType = 1 (see **Remarks**)

DWidth

Gusset width

DThickness

Gusset thickness

BDraft
True to draft gusset side faces, false to not
DDraftAngle
Angle by which to draft the gusset side faces; valid only if BDraft is true
BInnerCornerFillet
True to fillet the gusset inner corners, false to not
DInnerCornerFilletRadius
Inner corner fillet radius; valid only if BInnerCornerFillet is true
BOuterCornerFillet
True to fillet the gusset outer corners, false to not
DOuterCornerFilletRadius
Outer corner fillet radius; valid only if BOuterCornerFilletRadius is true
GussetType
Type of gusset as defined in [swSheetMetalRibGussetType_e](#)

BEdgeFillet
True to fillet the edges of a flat back gusset, false to not; valid only if GussetType is flat
DEdgeFilletRadius
Edge fillet radius for the flat back gusset; valid only if BEdgeFillet is true and GussetType is flat
ArrayOfFaces

Array containing one or two [IFace2s](#) that are the legs of the gusset feature:

- Two flat faces adjacent to a bend
- Two flat faces adjacent to a circular bend
- One cylindrical bend face
- One flat face and one non-planar face adjacent to a toroidal bend
- One toroidal bend face

(see **Remarks**)

ArrayRefLines
Array containing one of the following entities that is perpendicular to the gusset section plane:

- Linear [IEdge](#)
- [ISketchSegment](#)

(see **Remarks**)

ArrayRefPoints
Array containing one of the following to position the gusset section plane:

- [IVertex](#)
- [ISketchPoint](#)
- [IRefPoint](#)

(see **Remarks**)

Return Value

[IFeature](#)

 Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertSheetMetalGussetFeature2](#).

 Remarks

If you pass an empty array to...	Then...
ArrayRefLines	This method assumes the reference line is the bend line adjacent to the entities in ArrayOfFaces.
ArrayRefPoints	You must set BOffset to true and specify DOffset for an assumed reference point. For insertion of the first gusset, the assumed reference point is on the end of the sheet metal bend. For second and subsequent gusset insertions, the assumed reference point is on the edge of the gusset last inserted.

Instead of calling this method, you can call [IModelDocExtension::SelectByID2](#) to pre-select the references before calling [IFeatureManager::InsertSheetMetalGussetFeature](#).

If ProfDimType = 1, to dimension the gusset profile using...	Specify...
Both faces of the bend	• DLength of d1 leg • DHeight of d2 leg • BUseAngle = false • BFlipSides = true to swap d1 and d2 legs during dimensioning
One bend face and the angle it forms with the gusset	• DLength of d1 leg • DAngle of angle formed by gusset intersecting d1 leg • BUseAngle = true • BFlipSides = true to swap d1 and d2 legs during dimensioning
Opposite bend face and the angle it forms with the gusset	• DHeight of d2 leg • DAngle of angle formed by gusset intersecting d2 leg • BUseAngle = true • BFlipSides = true to swap d1 and d2 legs during dimensioning

See the SOLIDWORKS Help for more information about sheet metal gusset features.

 See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[ISMGussetFeatureData Interface](#)

 Availability

InsertSheetMetalGussetFeature3 Method (IFeatureManager)

[See Also](#) [Example](#)[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertSheetMetalGussetFeature3 Method (IFeatureManager)Obsolete. Superseded by [IFeatureManager::CreateDefinition](#) and [IFeatureManager::CreateFeature](#).

.NET Syntax

Visual Basic (Declaration)

```

Function InsertSheetMetalGussetFeature3( _
    ByVal BOffset As System.Boolean, _
    ByVal DOffset As System.Double, _
    ByVal BFlipOffsetSide As System.Boolean, _
    ByVal ProfDimType As System.Integer, _
    ByVal DIndentDepth As System.Double, _
    ByVal DLength As System.Double, _
    ByVal BUseAngle As System.Boolean, _
    ByVal DHeight As System.Double, _
    ByVal DAngle As System.Double, _
    ByVal BFlipSides As System.Boolean, _
    ByVal DWidth As System.Double, _
    ByVal DThickness As System.Double, _
    ByVal BDraft As System.Boolean, _
    ByVal DDraftAngle As System.Double, _
    ByVal BInnerCornerFillet As System.Boolean, _
    ByVal DInnerCornerFilletRadius As System.Double, _
    ByVal BOuterCornerFillet As System.Boolean, _
    ByVal DOuterCornerFilletRadius As System.Double, _
    ByVal GussetType As System.Integer, _
    ByVal BEdgeFillet As System.Boolean, _
    ByVal DEdgeFilletRadius As System.Double, _
    ByVal BOverrideDoc As System.Boolean, _
    ByVal BShowProfile As System.Boolean, _
    ByVal BShowCenter As System.Boolean _
) As Feature

```

Visual Basic (Usage)



```

Dim instance As IFeatureManager
Dim BOffset As System.Boolean
Dim DOffset As System.Double
Dim BFlipOffsetSide As System.Boolean
Dim ProfDimType As System.Integer
Dim DIndentDepth As System.Double
Dim DLength As System.Double
Dim BUseAngle As System.Boolean
Dim DHeight As System.Double
Dim DAngle As System.Double
Dim BFlipSides As System.Boolean
Dim DWidth As System.Double
Dim DThickness As System.Double
Dim BDraft As System.Boolean
Dim DDraftAngle As System.Double
Dim BInnerCornerFillet As System.Boolean
Dim DInnerCornerFilletRadius As System.Double
Dim BOuterCornerFillet As System.Boolean
Dim DOuterCornerFilletRadius As System.Double
Dim GussetType As System.Integer
Dim BEdgeFillet As System.Boolean
Dim DEdgeFilletRadius As System.Double
Dim BOverrideDoc As System.Boolean
Dim BShowProfile As System.Boolean
Dim BShowCenter As System.Boolean
Dim value As Feature

```

```
value = instance.InsertSheetMetalGussetFeature3(BOffset, DOffset, BFlipOffsetSide, ProfDimType,
DIndentDepth, DLength, BUseAngle, DHeight, DAngle, BFlipSides, DWidth, DThickness, BDraft,
DDraftAngle, BInnerCornerFillet, DInnerCornerFilletRadius, BOuterCornerFillet,
DOuterCornerFilletRadius, GussetType, BEdgeFillet, DEdgeFilletRadius, BOverrideDoc, BShowProfile,
BShowCenter)
```

C#

```
Feature InsertSheetMetalGussetFeature3(
    System.bool BOffset,
    System.double DOffset,
    System.bool BFlipOffsetSide,
    System.int ProfDimType,
    System.double DIndentDepth,
    System.double DLength,
    System.bool BUseAngle,
    System.double DHeight,
    System.double DAngle,
    System.bool BFlipSides,
    System.double DWidth,
    System.double DThickness,
    System.bool BDraft,
    System.double DDraftAngle,
    System.bool BInnerCornerFillet,
    System.double DInnerCornerFilletRadius,
    System.bool BOuterCornerFillet,
    System.double DOuterCornerFilletRadius,
    System.int GussetType,
    System.bool BEdgeFillet,
    System.double DEdgeFilletRadius,
    System.bool BOverrideDoc,
    System.bool BShowProfile,
    System.bool BShowCenter
)
```

C++/CLI

```
Feature^ InsertSheetMetalGussetFeature3(
& System.bool BOffset,
& System.double DOffset,
& System.bool BFlipOffsetSide,
& System.int ProfDimType,
& System.double DIndentDepth,
& System.double DLength,
& System.bool BUseAngle,
& System.double DHeight,
& System.double DAngle,
& System.bool BFlipSides,
& System.double DWidth,
& System.double DThickness,
& System.bool BDraft,
& System.double DDraftAngle,
& System.bool BInnerCornerFillet,
& System.double DInnerCornerFilletRadius,
& System.bool BOuterCornerFillet,
& System.double DOuterCornerFilletRadius,
& System.int GussetType,
& System.bool BEdgeFillet,
& System.double DEdgeFilletRadius,
& System.bool BOverrideDoc,
& System.bool BShowProfile,
& System.bool BShowCenter
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BOffset

True to offset the gusset section plane from the selected reference point, false to not

DOffset

Value by which to offset the gusset section plane from the selected reference point; valid only if BOffset is true

BFlipOffsetSide

True to specify the gusset section plane offset on the opposite side of the selected reference point, false to not; valid only if BOffset is true
ProfDimType

Type of gusset profile dimensioning scheme:

- 0 = indent
- 1 = profile dimensioning

(see **Remarks**)

DIndentDepth
Gusset depth; valid only if ProfDimType = 0

DLength
d1 gusset leg length; length of sheet metal from bend to intersection with first end of gusset; valid only if ProfDimType = 1 (see **Remarks**)

BUseAngle
True to dimension the angle that is formed by the intersection of the gusset with either face adjacent to the bend; valid only if ProfDimType = 1 (see **Remarks**)

DHeight
d2 gusset leg length; length of sheet metal from bend to intersection with second end of gusset; valid only if ProfDimType = 1 (see **Remarks**)

DAngle
Angle formed by intersection of gusset with one face adjacent to the bend; valid only if ProfDimType = 1 and BUseAngle is true (see **Remarks**)

BFlipSides
True to swap d1 and d2 in the dimensioning scheme, false to not; valid only if ProfDimType = 1 (see **Remarks**)

DWidth
Gusset width

DThickness
Gusset thickness

BDraft
True to draft gusset side faces, false to not

DDraftAngle
Angle by which to draft the gusset side faces; valid only if BDraft is true

BInnerCornerFillet
True to fillet the gusset inner corners, false to not

DInnerCornerFilletRadius
Inner corner fillet radius; valid only if BInnerCornerFillet is true

BOuterCornerFillet
True to fillet the gusset outer corners, false to not

DOuterCornerFilletRadius
Outer corner fillet radius; valid only if BOuterCornerFilletRadius is true

GussetType

Type of gusset:

- 0 = rounded back
- 1 = flat back

BEdgeFillet
True to fillet the edges of a gusset, false to not; valid only if GussetType = 1

DEdgeFilletRadius
Edge fillet radius; valid only if BEdgeFillet is true and GussetType = 1

BOVERRIDEdoc
True to override the document flat pattern properties, false to not

BShowProfile
True to display the gusset profile when flattened, false to not; valid only if BOVERRIDEdoc is true

BShowCenter
True to display the center marks when flattened, false to not; valid only if BOVERRIDEdoc is true

Return Value

[IFeature](#)

 Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertSheetMetalGussetFeature3](#).

 Example

[Insert Sheet Metal Gusset Feature \(VBA\)](#)
[Insert Sheet Metal Gusset Feature \(VB.NET\)](#)
[Insert Sheet Metal Gusset Feature \(C#\)](#)

 Remarks

Before calling this method to create a sheet metal gusset feature, you must pre-select faces, lines, edges, vertices, or points as follows:

To...	Call <code>IModelDocExtension::SelectByID2</code> to select...
Specify the legs of the gusset feature	- Or - • Two flat faces of a bend. - Or - • Two flat faces adjacent to a cylindrical bend face. - Or - • One cylindrical bend face. - Or - • One flat face and one non-planar face adjacent to a toroidal bend. - Or - • One toroidal bend face.
Optionally orient the gusset section plane	One linear edge or sketch segment that is perpendicular to the gusset section plane. If you do not pre-select either entity, then the method uses the bend line adjacent to the gusset legs as the reference.
Optionally position the gusset section plane	One vertex, sketch point, or reference point. If you do not pre-select any one of these entities, then you must set BOffset to true and specify DOffset for an assumed reference point. For insertion of the first gusset, the assumed reference point is on the end of the sheet metal bend. For second and subsequent gusset insertions, the assumed reference point is on the edge of the gusset last inserted.

If ProfDimType = 1, to dimension the gusset profile using...	Specify...
Both legs of the gusset	- DLength of d1 leg - DHeight of d2 leg - BUseAngle = false - BFlipSides = true to swap d1 and d2 legs during dimensioning
d1 leg and the angle it forms with the gusset	- DLength of d1 leg - DAngle of angle formed by gusset intersecting d1 leg - BUseAngle = true - BFlipSides = true to swap d1 and d2 legs during dimensioning
d2 leg and the angle it forms with the gusset	- DHeight of d2 leg - DAngle of angle formed by gusset intersecting d2 leg - BUseAngle = true - BFlipSides = true to swap d1 and d2 legs during dimensioning

See the SOLIDWORKS Help for more information about sheet metal gusset features.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[ISMGussetFeatureData Interface](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

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InsertSheetMetalHem Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertSheetMetalHem Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::InsertSheetMetalHem2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertSheetMetalHem( _
    ByVal Type As System.Integer, _
    ByVal Position As System.Integer, _
    ByVal Reverse As System.Boolean, _
    ByVal DLength As System.Double, _
    ByVal DGap As System.Double, _
    ByVal DAngle As System.Double, _
    ByVal DRad As System.Double, _
    ByVal DMiterGap As System.Double, _
    ByVal PCBA As CustomBendAllowance _
) As Feature
```

Visual Basic (Usage)

```
Dim instance As IFeatureManager
Dim Type As System.Integer
Dim Position As System.Integer
Dim Reverse As System.Boolean
Dim DLength As System.Double
Dim DGap As System.Double
Dim DAngle As System.Double
Dim DRad As System.Double
Dim DMiterGap As System.Double
Dim PCBA As CustomBendAllowance
Dim value As Feature
```

```
value = instance.InsertSheetMetalHem(Type, Position, Reverse, DLength, DGap, DAngle, DRad, DMiterGap, PCBA)
```

C#

```
Feature InsertSheetMetalHem(
    System.int Type,
    System.int Position,
    System.bool Reverse,
    System.double DLength,
    System.double DGap,
```

```
System.double DAngle,
System.double DRad,
System.double DMiterGap,
CustomBendAllowance PCBA
)
```

C++/CLI

```
Feature^ InsertSheetMetalHem(
& System.int Type,
& System.int Position,
& System.bool Reverse,
& System.double DLength,
& System.double DGap,
& System.double DAngle,
& System.double DRad,
& System.double DMiterGap,
& CustomBendAllowance^ PCBA
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Type

Type as defined in [swHemTypes_e](#)

Position

Position as defined in [swHemPositionTypes_e](#)

Reverse

True reverses the direction, false does not

DLength

Hem length; valid only for open or closed hems

DGap

Gap distance; valid only for open hems

DAngle

Hem angle; valid only for tear-drop or rolled hems

DRad

Hem radius; valid only for tear-drop or rolled hems

DMiterGap

Hem miter gap

PCBA

If...	Then...
non-NULL	Pointer to ICustomBendAllowance object for which required values have been set
NULL	Parent bend's bend allowance is used

Return Value

Pointer to the [IFeature](#) object

 Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertSheetMetalHem](#).

 See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IHemFeatureData Interface](#)
[ISheetMetalFeatureData Interface](#)

 Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0



Inserts a hem of the specified relief type at the selected edges of the current sheet metal part.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function InsertSheetMetalHem2( _  
    ByVal Type As System.Integer, _  
    ByVal Position As System.Integer, _  
    ByVal Reverse As System.Boolean, _  
    ByVal DLength As System.Double, _  
    ByVal DGap As System.Double, _  
    ByVal DAngle As System.Double, _  
    ByVal DRad As System.Double, _  
    ByVal DMiterGap As System.Double, _  
    ByVal PCBA As CustomBendAllowance, _  
    ByVal UseDefaultRelief As System.Boolean, _  
    ByVal ReliefType As System.Integer, _  
    ByVal ReliefTearTypes As System.Integer, _  
    ByVal UseReliefRatio As System.Boolean, _  
    ByVal ReliefRatio As System.Double, _  
    ByVal ReliefWidth As System.Double, _  
    ByVal ReliefDepth As System.Double _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Type As System.Integer  
Dim Position As System.Integer  
Dim Reverse As System.Boolean  
Dim DLength As System.Double  
Dim DGap As System.Double  
Dim DAngle As System.Double  
Dim DRad As System.Double  
Dim DMiterGap As System.Double  
Dim PCBA As CustomBendAllowance  
Dim UseDefaultRelief As System.Boolean  
Dim ReliefType As System.Integer  
Dim ReliefTearTypes As System.Integer  
Dim UseReliefRatio As System.Boolean  
Dim ReliefRatio As System.Double  
Dim ReliefWidth As System.Double  
Dim ReliefDepth As System.Double  
Dim value As Feature
```

```
value = instance.InsertSheetMetalHem2(Type, Position, Reverse, DLength, DGap, DAngle, DRad,  
DMiterGap, PCBA, UseDefaultRelief, ReliefType, ReliefTearTypes, UseReliefRatio, ReliefRatio,  
ReliefWidth, ReliefDepth)
```

C#

```
Feature InsertSheetMetalHem2(  
    System.int Type,  
    System.int Position,  
    System.bool Reverse,  
    System.double DLength,  
    System.double DGap,  
    System.double DAngle,  
    System.double DRad,  
    System.double DMiterGap,  
    CustomBendAllowance PCBA,  
    System.bool UseDefaultRelief,  
    System.int ReliefType,  
    System.int ReliefTearTypes,  
    System.bool UseReliefRatio,  
    System.double ReliefRatio,
```

```
System.double ReliefWidth,
System.double ReliefDepth
)
```

C++/CLI

```
Feature^ InsertSheetMetalHem2(
& System.int Type,
& System.int Position,
& System.bool Reverse,
& System.double DLength,
& System.double DGap,
& System.double DAngle,
& System.double DRad,
& System.double DMiterGap,
& CustomBendAllowance^ PCBA,
& System.bool UseDefaultRelief,
& System.int ReliefType,
& System.int ReliefTearTypes,
& System.bool UseReliefRatio,
& System.double ReliefRatio,
& System.double ReliefWidth,
& System.double ReliefDepth
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Type

Type as defined in [swHemTypes_e](#)

Position

Position as defined in [swHemPositionTypes_e](#)

Reverse

True reverses the direction, false does not

DLength

Hem length; valid only for open or closed hems

DGap

Gap distance; valid only for open hems

DAngle

Hem angle; valid only for tear-drop or rolled hems

DRad

Hem radius; valid only for tear-drop or rolled hems

DMiterGap

Hem miter gap

PCBA

If...	Then...
non-NULL	Pointer to ICustomBendAllowance object for which required values have been set
NULL	Parent bend's bend allowance is used

UseDefaultRelief

True to use the default relief, false to use the relief specified by ReliefType

ReliefType

Type of relief as defined in [swSheetMetalReliefTypes_e](#); valid only if UseDefaultRelief is false

ReliefTearTypes

Type of relief tear as defined in [swReliefTearTypes_e](#); valid only when:

- UseDefaultRelief is false
- and -
- ReliefType is [swSheetMetalReliefTypes_e](#).swSheetMetalReliefTear

UseReliefRatio

True to use ReliefRatio, false to use ReliefWidth/ReliefDepth; valid only when:

- UseDefaultRelief is false
- and -
- ReliefType is either [swSheetMetalReliefTypes_e](#).swSheetMetalReliefObround or [swSheetMetalReliefTypes_e](#).swSheetMetalReliefRectangular

ReliefRatio

Relief ratio; valid only when:

- UseDefaultRelief is false
- and -
- UseReliefRatio is true
- and -

ReliefType is either [swSheetMetalReliefTypes_e.swSheetMetalReliefObround](#) or [swSheetMetalReliefTypes_e.swSheetMetalReliefRectangular](#)

Width of the relief; valid only when:

- UseDefaultRelief is false
- and -
- UseReliefRatio is false
- and -

ReliefType is either [swSheetMetalReliefTypes_e.swSheetMetalReliefObround](#) or [swSheetMetalReliefTypes_e.swSheetMetalReliefRectangular](#)

Depth of the relief; valid only when:

- UseDefaultRelief is false
- and -
- UseReliefRatio is false
- and -

- ReliefType is either [swSheetMetalReliefTypes_e.swSheetMetalReliefObround](#) or [swSheetMetalReliefTypes_e.swSheetMetalReliefRectangular](#)

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertSheetMetalHem2](#).

Example

[Insert Sheet Metal Hem \(VBA\)](#)
[Insert Sheet Metal Hem \(VB.NET\)](#)
[Insert Sheet Metal Hem \(C#\)](#)

Remarks

Before calling this method:

1. Call [IFeatureManager::CreateCustomBendAllowance](#) to create an instance of ICustomBendAllowance.
2. Initialize the CustomBendAllowance object.
3. Assign PCBA to the initialized CustomBendAllowance object.
4. Select one or more edges in the sheet metal model to which to add the specified hem.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IHemFeatureData Interface](#)
[ISheetMetalFeatureData Interface](#)

Availability

SOLIDWORKS 2011 SP03, Revision Number 19.3

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertSheetMetalLoftedBend Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertSheetMetalLoftedBend Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::InsertSheetMetalLoftedBend2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertSheetMetalLoftedBend( _  
    ByVal ThickDirType As System.Integer, _  
    ByVal Thickness As System.Double _  
    ) As Feature
```

Visual Basic (Usage)

```
Dim instance As IFeatureManager  
Dim ThickDirType As System.Integer
```



```
Dim Thickness As System.Double
Dim value As Feature

value = instance.InsertSheetMetalLoftedBend(ThickDirType, Thickness)
```

C#

```
Feature InsertSheetMetalLoftedBend(
    System.int ThickDirType,
    System.double Thickness
)
```

C++/CLI

```
Feature^ InsertSheetMetalLoftedBend(
    & System.int ThickDirType,
    & System.double Thickness
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ThickDirType

Toggles the thickening direction:

- 0 = outside
- 1 = inside (default value)

Thickness

Thickness of the bend

Return Value

Pointer to the lofted-bend [feature](#) in the sheet metal part

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertSheetMetalLoftedBend](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[ILoftFeatureData Interface](#)
[ISheetMetalFeatureData Interface](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertSheetMetalLoftedBend2 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertSheetMetalLoftedBend2 Method (IFeatureManager)

Inserts a lofted bend in a sheet metal part.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertSheetMetalLoftedBend2( _
    ByVal ThickDirType As System.Integer, _
    ByVal Thickness As System.Double, _
    ByVal BFormed As System.Boolean, _
    ByVal DRadius As System.Double, _
    ByVal BReferToEndPoint As System.Boolean, _
    ByVal EFacetOption As System.Integer, _
    ByVal DChordTol As System.Double, _
    ByVal INumBends As System.Integer, _
```

```

    ByVal DSegLength As System.Double, _
    ByVal DSegAngle As System.Double _
) As Feature

```

Visual Basic (Usage)



```

Dim instance As IFeatureManager
Dim ThickDirType As System.Integer
Dim Thickness As System.Double
Dim BFormed As System.Boolean
Dim DRadius As System.Double
Dim BReferToEndPoint As System.Boolean
Dim EFacetOption As System.Integer
Dim DChordTol As System.Double
Dim INumBends As System.Integer
Dim DSegLength As System.Double
Dim DSegAngle As System.Double
Dim value As Feature

value = instance.InsertSheetMetalLoftedBend2(ThickDirType, Thickness, BFormed, DRadius,
BReferToEndPoint, EFacetOption, DChordTol, INumBends, DSegLength, DSegAngle)

```

C#

```

Feature InsertSheetMetalLoftedBend2(
    System.int ThickDirType,
    System.double Thickness,
    System.bool BFormed,
    System.double DRadius,
    System.bool BReferToEndPoint,
    System.int EFacetOption,
    System.double DChordTol,
    System.int INumBends,
    System.double DSegLength,
    System.double DSegAngle
)

```

C++/CLI

```

Feature^ InsertSheetMetalLoftedBend2(
& System.int ThickDirType,
& System.double Thickness,
& System.bool BFormed,
& System.double DRadius,
& System.bool BReferToEndPoint,
& System.int EFacetOption,
& System.double DChordTol,
& System.int INumBends,
& System.double DSegLength,
& System.double DSegAngle
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ThickDirType

Toggles the thickening direction:

- 0 = outside
- 1 = inside (default value)

Thickness

Thickness of the bend

BFormed

True to insert a formed lofted bend; false to insert a bent lofted bend (default)

DRadius

Bend radius; valid only if BFormed is false

BReferToEndPoint

True to calculate facet transitions using theoretical vertexes, false to calculate facet transitions using the smallest possible arcs that avoid interference between bend areas; valid only if BFormed is false

EFacetOption

Faceting option as defined in [swLoftedBendFacetOptions](#) e; valid only if BFormed is false

DChordTol
Chord tolerance or maximum distance between the arc and linear segment of a chord; valid only if EFacetOption is set to swLoftedBendFacetOptions_e.swChordTolerance and BFormed is false
INumBends
Maximum number of bends per transition segment; valid only if EFacetOption is set to swLoftedBendFacetOptions_e.swBendsPerTransitionSegment and BFormed is false
DSegLength
Maximum length of the linear segment; valid only if EFacetOption is set to swLoftedBendFacetOptions_e.swSegmentLength and BFormed is false
DSegAngle
Maximum angle between adjacent linear segments; valid only if EFacetOption is set to swLoftedBendFacetOptions_e.swSegmentAngle and BFormed is false

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertSheetMetalLoftedBend2](#).

Example

[Insert Lofted Bend Feature \(VBA\)](#)
[Insert Lofted Bend Feature \(VB.NET\)](#)
[Insert Lofted Bend Feature \(C#\)](#)

Remarks

Before calling this method, you must select at least two sketch profiles that are on parallel planes. You can either multi-select them in the graphics area or call [IModelDocExtension::SelectById2](#) with Mark = 1 for each sketch profile you want to select.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[ILoftedBendsFeatureData Interface](#)
[ILoftFeatureData Interface](#)
[ISheetMetalFeatureData Interface](#)

Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertSheetMetalMiterFlange Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertSheetMetalMiterFlange Method (IFeatureManager)

Inserts a meter flange in a sheet metal part.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertSheetMetalMiterFlange( _  
    ByVal UseDefaultRadius As System.Boolean, _  
    ByVal GlobalRadius As System.Double, _  
    ByVal RipGap As System.Double, _  
    ByVal UseDefaultRelief As System.Boolean, _  
    ByVal UseReliefRatio As System.Boolean, _  
    ByVal ReliefRatio As System.Double, _  
    ByVal ReliefWidth As System.Double, _  
    ByVal ReliefDepth As System.Double, _  
    ByVal ReliefType As System.Integer, _  
    ByVal TrimSideBends As System.Boolean, _  
    ByVal FlangePos As System.Integer, _  
    ByVal OffsetDist1 As System.Double, _  
    ByVal OffsetDist2 As System.Double, _  
    ByVal PCBA As CustomBendAllowance _  
) As Feature
```

Visual Basic (Usage)



```

Dim instance As IFeatureManager
Dim UseDefaultRadius As System.Boolean
Dim GlobalRadius As System.Double
Dim RipGap As System.Double
Dim UseDefaultRelief As System.Boolean
Dim UseReliefRatio As System.Boolean
Dim ReliefRatio As System.Double
Dim ReliefWidth As System.Double
Dim ReliefDepth As System.Double
Dim ReliefType As System.Integer
Dim TrimSideBends As System.Boolean
Dim FlangePos As System.Integer
Dim OffsetDist1 As System.Double
Dim OffsetDist2 As System.Double
Dim PCBA As CustomBendAllowance
Dim value As Feature

```

```

value = instance.InsertSheetMetalMiterFlange(UseDefaultRadius, GlobalRadius, RipGap,
UseDefaultRelief, UseReliefRatio, ReliefRatio, ReliefWidth, ReliefDepth, ReliefType,
TrimSideBends, FlangePos, OffsetDist1, OffsetDist2, PCBA)

```

C#

```

Feature InsertSheetMetalMiterFlange(
    System.bool UseDefaultRadius,
    System.double GlobalRadius,
    System.double RipGap,
    System.bool UseDefaultRelief,
    System.bool UseReliefRatio,
    System.double ReliefRatio,
    System.double ReliefWidth,
    System.double ReliefDepth,
    System.int ReliefType,
    System.bool TrimSideBends,
    System.int FlangePos,
    System.double OffsetDist1,
    System.double OffsetDist2,
    CustomBendAllowance PCBA
)

```

C++/CLI

```

Feature^ InsertSheetMetalMiterFlange(
    & System.bool UseDefaultRadius,
    & System.double GlobalRadius,
    & System.double RipGap,
    & System.bool UseDefaultRelief,
    & System.bool UseReliefRatio,
    & System.double ReliefRatio,
    & System.double ReliefWidth,
    & System.double ReliefDepth,
    & System.int ReliefType,
    & System.bool TrimSideBends,
    & System.int FlangePos,
    & System.double OffsetDist1,
    & System.double OffsetDist2,
    CustomBendAllowance^ PCBA
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UseDefaultRadius
True to use default bend radius, false to use GlobalRadius
GlobalRadius
Global bend radius

RipGap
Rip-gap distance

UseDefaultRelief
True to use parent bend's ratio, false to not

UseReliefRatio
True to use custom relief ratio, false to not
ReliefRatio
If UseReliefRatio is True, then specify relief ratio
ReliefWidth
If UseReliefRatio is True and ReliefType is swSheetMetalReliefRectangular or swSheetMetalReliefObround, then specify relief depth

ReliefDepth
If UseReliefRatio is True and ReliefType is swSheetMetalReliefRectangular or swSheetMetalReliefObround, then specify relief depth

- ReliefType*
Relief type as defined by [swSheetMetalReliefTypes_e](#); valid selections:
- swSheetMetalReliefObround
 - swSheetMetalReliefRectangular
 - swSheetMetalReliefTear

TrimSideBends
True to trim side bends, false to not

FlangePos
Flange position as defined by [swFlangePositionTypes_e](#); valid selections:

- swFlangePositionTypeMaterialInside
- swFlangePositionTypeMaterialOutside
- swFlangePositionTypeBendOutside

OffsetDist1
Starting offset distance if partial flange
OffsetDist2
Ending offset distance if partial flange
PCBA

If...	Then...
non-NULL	Pointer to ICustomBendAllowance object for which required values have been set
NULL	Parent bend's bend allowance is used

Return Value

Pointer to the [IFeature](#) object or NULL

 Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertSheetMetalMiterFlange](#).

 See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IMiterFlangeFeatureData Interface](#)
[ISheetMetalFeatureData Interface](#)

 Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

InsertSlicing Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertSlicing Method (IFeatureManager)

Creates and inserts slicing into the FeatureManager design tree.

 .NET Syntax

Visual Basic (Declaration)

```
Function InsertSlicing( _  
    ByVal SlicingData As System.Object, _  
    ByRef Errors As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```

Dim instance As IFeatureManager
Dim SlicingData As System.Object
Dim Errors As System.Integer
Dim value As System.Object

value = instance.InsertSlicing(SlicingData, Errors)

```

C#

```

System.object InsertSlicing(
    System.object SlicingData,
    out System.int Errors
)

```

C++/CLI

```

System.Object^ InsertSlicing(
    & System.Object^ SlicingData,
    & [Out] System.int Errors
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SlicingData
[ISlicingData](#) (see **Remarks**)
Errors
 Errors as defined in [swInsertSlicingError_e](#)

Return Value

Array of sketch and reference plane objects

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertSlicing](#).

Example

See the [ISlicingData](#) example.

Remarks

This method invokes the Slicing tool. For more information, refer to **SOLIDWORKS Help > Parts and Features > Controlling Parts > Slicing Tool**.

Before calling this method:

1. Pre-select in the graphics area a planar entity (to create a linear pattern of slices) or a combination of a linear entity and a point entity (to create an angular pattern of slices whose axis is the linear entity).
2. Use [IFeatureManager::GetSlicingData](#) to get an ISlicingData object.
3. Set [ISlicingData::PlaneReferences](#) if step 1 is not performed. Specify a planar entity or a combination of a linear entity and a point entity as specified in step 1.
4. Set other ISlicingData properties.
5. Populate SlicingData with the ISlicingData object.

After calling this method:

- Use the array of sketch and reference plane objects returned by this method to perform further work.
- If [ISlicingData::AddSlicingPlanesAndSketchesToFolder](#) was set to true, then a **Slice1** folder in the FeatureManager design tree is created containing the slicing planes and sketches. You can edit the slicing planes and sketches individually as needed.
- Delete the Slice1 folder and its contents to remove slicing from the model.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertSplitLineIntersect Method (IFeatureManager)

Creates split lines using the selected intersecting tool and target entities.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertSplitLineIntersect(_  
    ByVal CompleteOption As System.Integer _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim CompleteOption As System.Integer  
Dim value As Feature
```

```
value = instance.InsertSplitLineIntersect(CompleteOption)
```

C#

```
Feature InsertSplitLineIntersect(  
    System.int CompleteOption  
)
```

C++/CLI

```
Feature^ InsertSplitLineIntersect(  
& System.int CompleteOption  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

CompleteOption

- 1 = Linear
- 7 = Natural
- 9 = Split all and linear
- 15 = Split all and natural

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertSplitLineIntersect](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[ISplitLineFeatureData Interface](#)
[IModelDoc2.InsertSplitLineProject Method](#)
[IModelDoc2.InsertSplitLineSil Method](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertStructuralWeldment Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertStructuralWeldment Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager.InsertStructuralWeldment2](#).

.NET Syntax

```
Function InsertStructuralWeldment( _
    ByVal Path As System.String, _
    ByVal EndCond As System.Integer, _
    ByVal Angle As System.Double _
) As Feature
```



```
Dim instance As IFeatureManager
Dim Path As System.String
Dim EndCond As System.Integer
Dim Angle As System.Double
Dim value As Feature

value = instance.InsertStructuralWeldment(Path, EndCond, Angle)
```

```
Feature InsertStructuralWeldment(
    System.string Path,
    System.int EndCond,
    System.double Angle
)
```

```
Feature^ InsertStructuralWeldment(
    & System.String^ Path,
    & System.int EndCond,
    & System.double Angle
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Path

EndCond

Angle

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertStructuralWeldment](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

InsertStructuralWeldment2 Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertStructuralWeldment2 Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::InsertStructuralWeldment3](#).

.NET Syntax

```
Function InsertStructuralWeldment2( _
    ByVal Path As System.String, _
    ByVal EndCond As System.Integer, _
    ByVal Angle As System.Double, _
    ByVal Merge As System.Boolean _
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager
Dim Path As System.String
Dim EndCond As System.Integer
Dim Angle As System.Double
Dim Merge As System.Boolean
Dim value As Feature

value = instance.InsertStructuralWeldment2(Path, EndCond, Angle, Merge)
```

C#

```
Feature InsertStructuralWeldment2(
    System.string Path,
    System.int EndCond,
    System.double Angle,
    System.bool Merge
)
```

C++/CLI

```
Feature^ InsertStructuralWeldment2(
    & System.String^ Path,
    & System.int EndCond,
    & System.double Angle,
    & System.bool Merge
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Path

Path, filename, and name of the type of structural member to insert

EndCond

End condition as defined in [swSolidworksWeldmentEndCondOptions_e](#)

Angle

Angle of rotation of the sketch about the sketch segment

Merge

True to merge the bodies of the arc segments to the adjacent bodies, false to not

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertStructuralWeldment2](#).

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

[IStructuralMemberFeatureData Interface](#)

[IFeatureManager.InsertStructuralWeldment3 Method](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0